CHAPTER

3

Dental Caries

Nature has provided teeth to perform the functions of cutting, grinding and admixing food with saliva. The hard enamel cover along with periodontal ligament can withstand the forces of mastication. The unique character of tooth is that it is comprised of two tissues of different embryological origin; ectodermal enamel, and mesodermal dentin. Enamel is acellular, avascular and has no nerve supply. It is incapable of any natural defense mechanism based on cellular activity. As the tooth erupts, it is open to complex environment.

Dental caries is the most prevalent disease of mankind. It is strange that the hardest tissue of body, the enamel, which is indestructible otherwise, can disintegrate in the oral environment. Caries (Latin meaning 'dry rot') implies slow disintegration that may affect any of the biological hard tissue as a result of bacterial action. This action may affect bone, causing 'bone caries'. Usually such disintegration affects enamel, dentin and cementum (the tooth); that is why the term 'dental caries' is most common.

The pattern of dental caries is not uniform. This may be because of several important variables that affect caries, such as diet and other related factors. Different types of caries may have the same etiology; however, the susceptibility of different surfaces of teeth varies. The attack of caries and its severity also varies from individual-to-individual.

DEFINITIONS

In simple terms, dental caries is described as the irreversible, slow progressing decay of hard tissues of the tooth. It is a local disease, which involves destruction of hard tissues of the teeth by metabolites produced by oral micro-organisms.

It is defined as *"the microbial disease of the calcified tissues of teeth, characterized by demineralization of the inorganic portion and destruction of organic substance of the tooth"*.

According to WHO, caries is defined as *"a localized post-eruptive, pathological process of external origin involving softening of the hard tooth tissue and proceeding to the formation of a cavity"*.

CLASSIFICATION OF CARIES

Dental caries can be classified according to three major factors:

A. According to morphology of teeth
B. According to severity and progress of lesion
C. According to age

A. According to Morphology of Teeth

According to morphology of teeth, the caries is categorized into following four types.

a. ***Pit and fissure caries:*** Caries occurring on anatomical pits and fissures of all the teeth. This is also referred to as Class I caries (Figs 3.1 and 3.2A, B).

b. ***Smooth surface caries:*** Caries occurring on smooth surfaces of the teeth (Figs 3.1, 3.3A, B, and 3.4A, B).

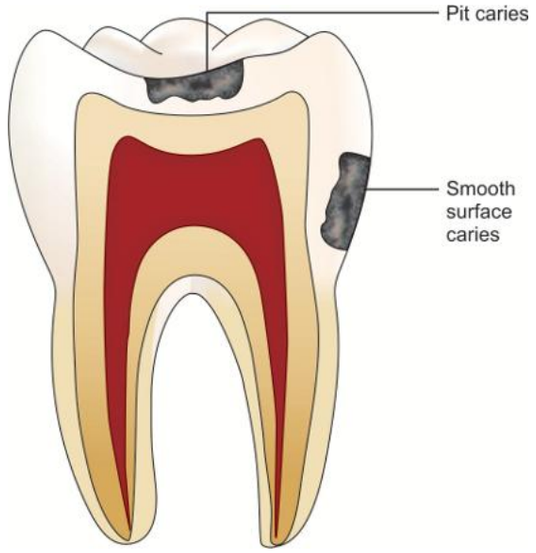


Fig. 3.1: Caries in the pits and fissures and on smooth surface (diagrammatic)

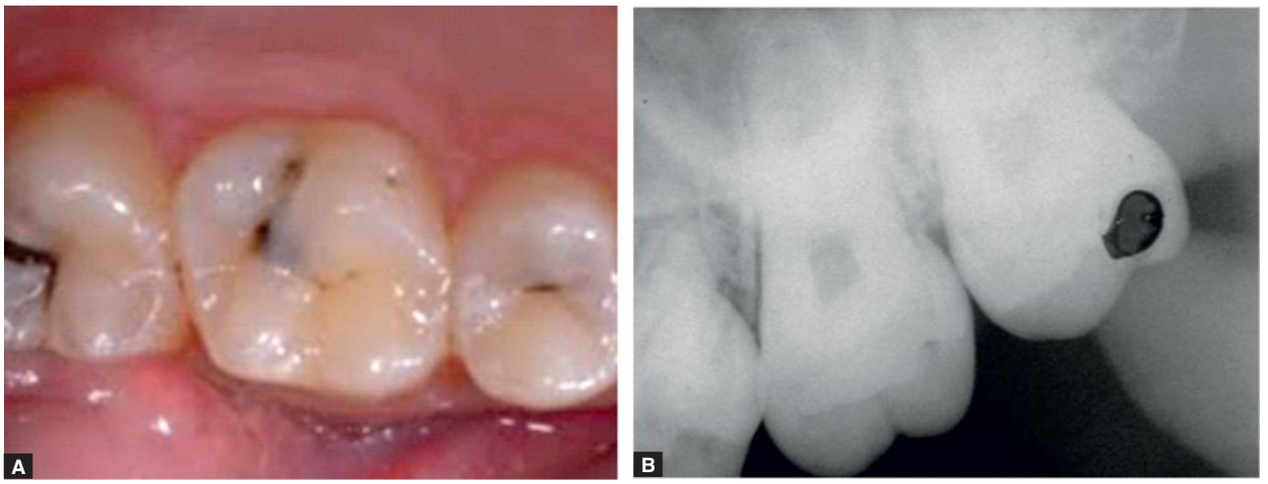


Fig. 3.2: Caries in the pit and fissure: **(A)** Clinical; **(B)** Radiographic

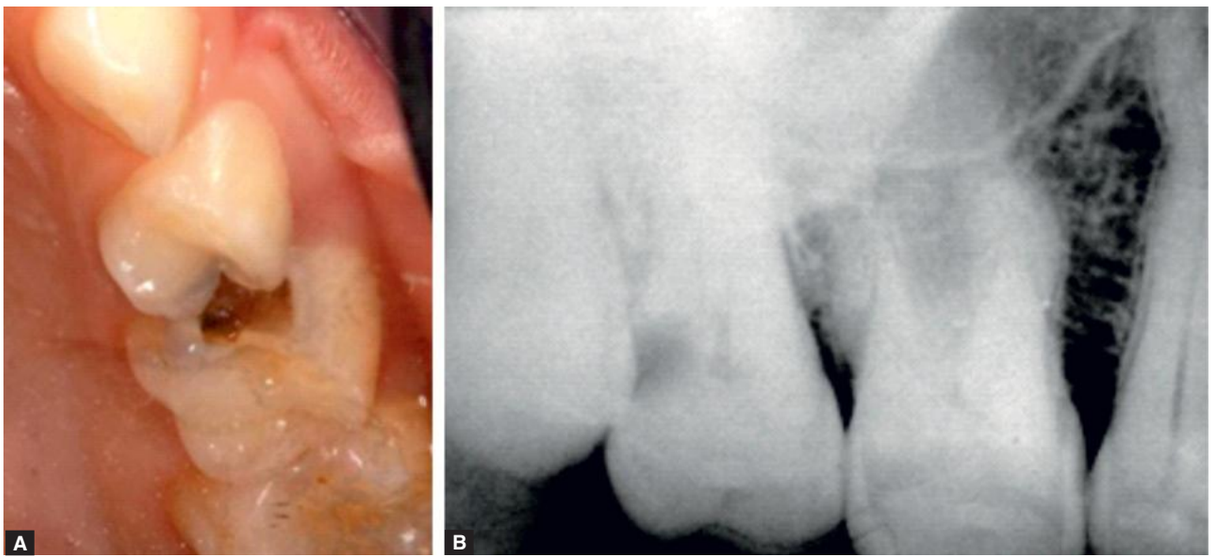


Fig. 3.3: Caries on smooth surface: **(A)** Clinical; **(B)** Radiographic

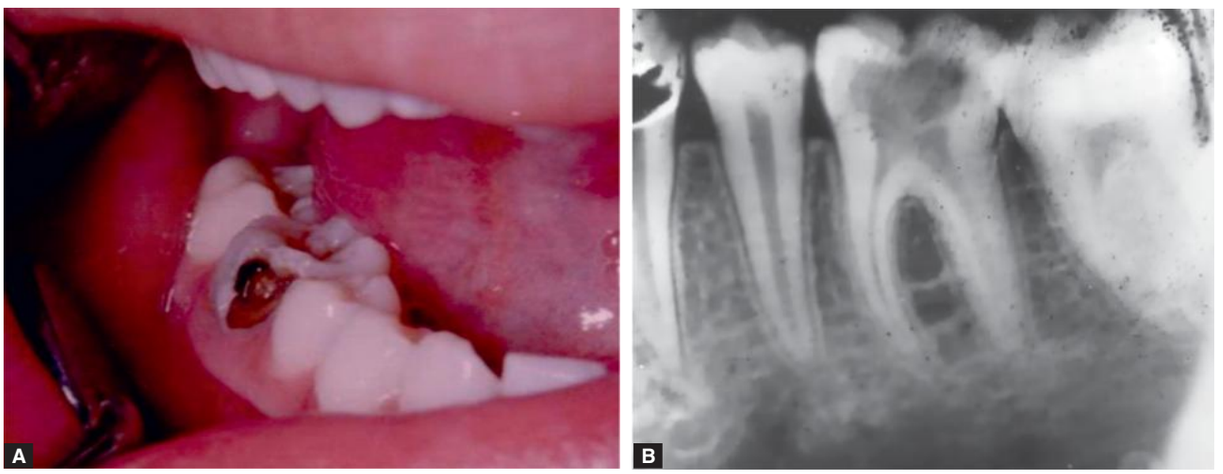


Fig. 3.4: Caries on smooth surface—buccal surface: **(A)** Clinical; **(B)** Radiographic

c. ***Root caries:*** Caries developing at the cementoenamel junction or cementum. This occurs predominantly in the older age when there is gingival recession (Figs 3.5 to 3.7).
d. ***Linear enamel caries:*** Caries developing on the labial surfaces of anterior teeth. This is also known as 'odontoclasia'. Such caries at the neonatal zone is because of trauma at birth or metabolic disturbances (Fig. 3.8).

B. According to Severity and Progress of Lesion

According to severity and progression of lesion, the caries is categorized into following five types.

a. ***Incipient caries:*** Incipient caries appears as a white opaque region on any tooth surface (Fig. 3.9). The white opacity is because of surface demineralization of enamel. The incipient lesion can undergo remineralization thereby reversing the process.

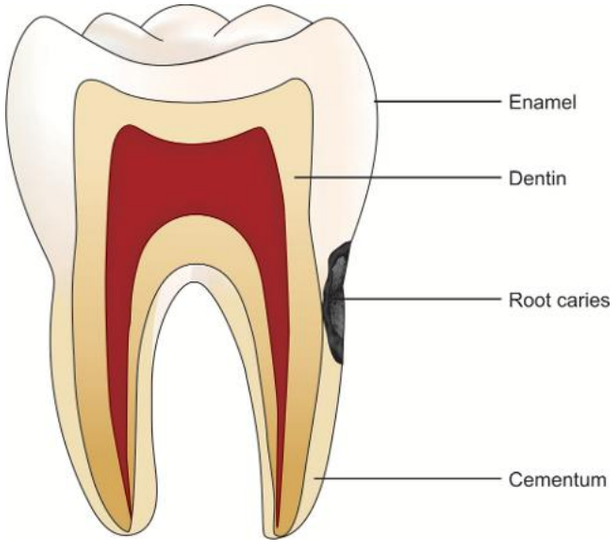


Fig. 3.5: Root caries (diagrammatic)



Fig. 3.6: Root caries (clinical)

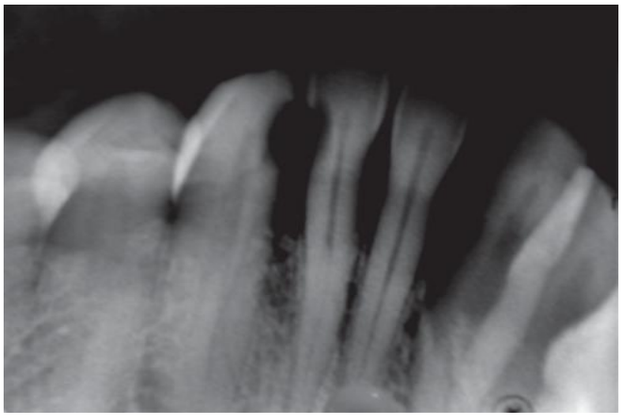


Fig. 3.7: Caries on the cementoenamel junction (radiographic)



Fig. 3.8: Linear enamel caries



Fig. 3.9: Incipient caries



Fig. 3.10: Rampant caries



Fig. 3.12: Recurrent (secondary) caries

b. ***Rampant caries:*** It is the sudden and rapid onset of caries involving at least two teeth and two surfaces (Fig. 3.10). A caries increment of ten or more new carious lesions over a year is characteristic of rampant caries.
c. ***Arrested caries:*** Any carious lesion, may get arrested, if there is change in the oral environment. The arrested caries, clinically appears as a dark brown pigmentation with smooth surface (Fig. 3.11). This is also referred to as 'Eburnation', derived from Latin word, which means arrested caries.
d. ***Recurrent (secondary) caries:*** Caries developing at the interface of tooth and restorative material is referred to as recurrent (secondary) caries (Fig. 3.12). It may be due to defective cavity preparation/restoration leading to microleakage.
e. ***Radiation caries:*** The radiotherapy of oral cancer lesions may lead to xerostomia, subsequently development of caries, known as radiation caries (Fig. 3.13).



Fig. 3.13: Radiation caries



Fig. 3.11: Arrested caries

C. According to Age

According to age, the caries is categorized into following three types.

a. ***Nursing bottle caries:*** In early infancy period, bottle fed babies develop caries, usually on maxillary incisors. The prolonged breastfeeding, especially at night, may result in such caries.
b. ***Adolescent caries:*** Caries attack during adolescent period is characterized as adolescent caries.
c. ***Geriatric caries:*** Caries developing in older adults is referred to as geriatric caries. Usually caries of cementum falls under this category.

ETIOLOGY OF DENTAL CARIES

Dental caries is a multifactorial disease. Till today, no single theory can explain the phenomenon of caries. The caries does not fall into any of the pathological lesions of oral cavity. The caries is not inflammatory in origin nor is it degenerative in nature and neither it is a neoplasm. Dental caries is a local disease, which involves destruction of hard tissues of tooth by metabolites produced by micro-organisms.

Evidence for caries has been found in Homo sapiens since Palaeolithic times. Numerous references have been documented to explain the caries process.

The possible theories are:

a. Worm Theory

The earliest reference of tooth decay appeared around 14th century BC. According to earlier concept, the cause of caries was thought to be invasion of 'worms' into teeth. The association of systemic disease and teeth was documented in the writings of a physician around 668 BC, who mentioned that the inflammation in his arms and legs was due to tooth and that it must be extracted.

b. Humoral Theory

Galen hypothesized that dental caries was produced by the action of acids along with corroding humors. The four recognized humors of the body were blood, phlegm, black bile and yellow bile. The imbalance in these humors resulted in the disease process. A few authors (Hippocrates) favoured this concept and added that accumulated debris around the teeth helped in corroding action.

c. Chemical Theory

Robertson (1835) proposed that acids, formed by fermentation of food particles, cause dental decay. The exact nature of acids and the mechanism of their formation were not known. Different postulates were suggested. One, putrefaction of protein gave rise to ammonia, which was subsequently oxidized to nitric acid. Another postulate was that the food was decomposed to sulphuric acid. Till then, the activity of bacteria was not recognized.

d. Parasite Theory

Erdl (1843) described filamentous parasites in the membrane of tooth surfaces. Dental caries was thought to develop as a result of infiltration of parasites and decomposition of enamel cuticle (surface protein membrane on the enamel).

e. Acidogenic Theory

Miller (1889) propagated the concept of acid formation in the oral cavity by the action of micro-organisms. He was of the view that micro-organisms of the oral cavity, by secretion of enzymes or by their own metabolites degrade the carbohydrates sticking over the tooth surfaces into acids. The acids formed were recognized as lactic acid, butyric acid, etc. The acid demineralized the enamel surface. After the disintegration of enamel, the organisms along with acids penetrate dentinal tubules leading to dissolution of dentin. The proteolytic enzymes finally digest the organic part.

The three factors recognized in Miller's observation are:

i. Micro-organisms responsible for the caries process.
ii. Carbohydrate substrate over the tooth surface.
iii. Production of acids initially and protein degradation subsequently.

He summarized his theory as "*Dental decay is a chemicoparasitic process consisting of the dissolution of the hard inorganic and the soft organic part*". In case of enamel, since it consists mainly of inorganic constituents, the second stage is missing.

A few authors presented experimental evidence implicating both acids and bacteria as the causative agents of caries. A specific micro-organism, *Leptothrix buccalis* was observed in dentinal tubules, giving the idea that the acids and micro-organisms collectively dissolved the organic part.

Miller was of the view that no single species of micro-organisms were capable of producing acids and digesting proteins. His work was confirmed by the following facts:

- Acid was present in the deeper carious lesions.
- Several micro-organisms of oral cavity were capable of producing acids.
- Lactic acid was identified in carbohydrate–saliva mixtures.
- Different micro-organisms had the potential to invade enamel and dentin.

Drawbacks

Though Miller's theory is relevant; however, following points were not clear and need explanation:

- Caries on the smooth surfaces of teeth.
- Particular type of micro-organism could not be isolated, which is responsible for particular acid.
- Phenomenon of arrested caries.
- Certain populations exhibit less caries despite of consuming enough carbohydrates.
- Caries of un-erupted/impacted teeth.
- Carious dentin, if left under a filling, continues to decay.

f. Proteolytic Theory

Gottlieb (1944) was of the view that instead of decalcification of inorganic part, as suggested in Miller's theory, the initial action is due to the proteolytic enzymes attacking the organic components, such as, lamellae, rod sheaths, tufts and walls of tubules, etc. The yellow pigmentation, which he observed with dental caries was attributed to the pigments produced by proteolytic organisms. Caries

is initiated at a slightly alkaline pH produced by the proteolytic activity liquefying the organic matrix of enamel. Once the inorganic part sets free after the dissolution of organic part, these salts are dissolved subsequently by acidogenic bacteria. He was of the view that the staphylococci play a vital role in initiating proteolytic activity.

Later, *Pincus* (1949) also observed that the initial process in caries was the proteolytic breakdown of the dental cuticle, followed by destruction of the prism sheaths. The loosened prisms then fell out mechanically. He proposed that the Nasmyth's membrane and the enamel proteins (mucoproteins) are acted upon by the sulphatase enzyme of the micro-organisms, producing sulphuric acid.

Drawbacks

- Enamel is a highly mineralized tissue. Though enamel contains 1.0 to 1.5% organic matrix out of which 0.6% is protein, initiation of caries with breakdown of this small percentage of protein is highly questionable.
- The sulfatase enzymes of gram-negative bacilli, which are considered to dissolve the mucoitin sulfate of enamel, have not been found in abundance in experimental studies. Moreover, enzyme systems capable of attacking keratin have not been demonstrated.
- This theory lacks experimental support.

g. Proteolysis Chelation Theory

Schatz et al. (1955) hypothesized that the microbial degradation of organic component by proteolysis followed by dissolution of inorganic part by the process of chelation lead to caries. The word 'chelate' refers to compounds that are able to bind metallic ions such as calcium, iron, copper, zinc, etc. by valence bonds. The proteolysis chelation theory considers dental caries to be bacterial destruction of organic component of enamel and the breakdown products of these organic components to have chelating properties and thereby dissolve the minerals in the enamel even at the neutral/alkaline pH. A variety of agents, such as amino acids, amines and peptides, etc., are the breakdown products of organic components of enamel and dentin, which can act as chelates.

Drawbacks

- Since the enamel contains very little amount of organic component, can organic dissolution produce sufficient amount of chelates to disintegrate the rest of inorganic enamel?
- Although chelation is an accepted biological phenomenon, its role in caries initiation has not been established.

The analysis of evidence supporting the three main theories of dental caries: Acidogenic theory, proteolysis theory and proteolysis chelation theory is summarized in Table 3.1.

h. Bioelectric Phenomenon

The mechanisms involved in production of electric voltages are basically piezoelectric effect (voltage produced by physical deformation of bone crystals), ion pump and membrane potential (involved in nerve activity), fuel cells (difference in electron concentrations where high and low redox potentials exist in close proximity with each other). Enamel surfaces of teeth and adjacent structures could develop electrical voltages by one or all the following: Bony crystalline structures, enamel that may act as ion permeable membrane and an intensely layered bacterial system (the plaque).

The reduction of pH by bacterial activity coupled with the voltage produced is responsible for inducing demineralization of tooth enamel.

Dental plaque may act as an electrical fuel-cell. The back of the plaque (tooth side) is found to be more electronegative as compared to the front. The addition of small amounts of glucose solution stimulates the front of this bacterial mat, which becomes electro-negative. As substrate penetrates past the surface, there is concurrent shift in potential, until the back and front of the bacterial mat are at the same negativity till finally the tooth region of the plaque becomes strongly electronegative with respect to the front. The electrical energies on the order of those developed by plaque will decalcify sound tooth structure. This electrical activity is stimulated by the presence of sugar.

The decalcification occurring because of electrical energies does indicate that there might be bioelectric phenomena involved in dental caries.

i. Ionic See-Saw (Levine's) Theory

Levine (1977) hypothesized 'see-saw' mechanism in which there is movement of minerals from saliva/plaque to enamel and vice versa (Fig. 3.14). The mechanism emphasized that the demineralization and remineralization of enamel is a continuous process. If in a given interval of time, more ions leave the enamel than enter, then there is a net demineralization, which amounts to the start of the carious process. It has been proved that the passage of ions is not a one way process and that ions are constantly being exchanged between enamel and plaque. At times, the

Table 3.1: Summary of analysis of evidence supporting three main theories of dental caries

Factors studied	*Acidogenic*	*Proteolysis*	*Chelation*
A. Clinical			
i. Site of caries (caries developing at places where fermentable carbohydrate food is retained on tooth surfaces)	+ The localization of caries is confirmed by various authors supported by clinical experience	– No evidence. Only proteolysis would account for localization, since carbohydrates are used for bacterial energy in preference to protein. Proteolysis will be delayed in areas where carbohydrates are retained	– No evidence as to why chelation account for localization
ii. Diet and caries (High incidence of caries observed in persons consuming high carbohydrate diet)	+ Clinical studies have established the relationship of caries and carbohydrates. Elimination of diet, which forms acids in oral cavity has been shown to reduce caries. Fluorides increase the resistance of enamel to decalcification to acids; hence, reduce caries	– No similar association exists in relation to attack by proteolysis. Increase resistance with fluorides not approved	– No similar association exists in relation to chelation
iii. Animal studies (Caries detected in animals when fed on carbohydrate diets and prevented by agents which inhibit fermentation)	+ Agents such as iodoacetic acid and dicalcium phosphate reduce caries in animals	– Evidence documented are questionable	– No evidence
B. Mechanism of destruction			
i. Destruction of inorganic part has been confirmed both *in vivo* and *in vitro*	+ (Decalcification by acid production has been proved)	± (No mechanism has been established to show how proteolysis will destroy the calcified tissues)	± (The keratin in enamel is dissolved by oral organisms, thereby forming chelating agents, which dissolve enamel. Evidence reported that the total keratin changed into chelating compounds can dissolve only 1.0% of calcium in enamel, so not convincing evidence to produce detectable caries)
ii. Destruction of organic part has doubtful evidence in all the three theories	± (Since organic portion is fragile, it disintegrates as decalcification takes place)	± (Proteolytic agents destroy human enamel protein is partially supported by scientific evidence)	± (It was presumed that oxygen uptake was observed when proteolytic bacterium grew in human enamel. But actual organisms were not isolated from enamel. The uptake of oxygen by enamel was as good as autorespiration)

(*Contd.*)

Table 3.1: Summary of analysis of evidence supporting three main theories of dental caries (*Contd.*)

Factors studied	*Acidogenic*	*Proteolysis*	*Chelation*
C. Histology of caries			
i. Destruction of enamel is accompanied by loss of calcific material; the organic portion persists	+ (Different zones of caries in enamel can be explained by acid theory)	± (Bacteria penetrate along the organic tracts of enamel; proteolysis does not explain different zones)	– (No evidence)
ii. Appearance of radiolucent area in X-rays	+ (Radiolucency seen in radiographs because of loss of calcium)	– (Support/evidence lacking)	– (Support/evidence lacking)
iii. Increase of nitrogen and loss of specific gravity and hardness	+ (Loss of inorganic material and persistence of organic material occur in initial stage which account for increase of nitrogen and loss of specific gravity and hardness)	– (No evidence)	– (No evidence)
D. Bacteriology			
i. Isolation of appropriate bacteria from enamel caries	+ (Bacteria producing acids from fermentable carbohydrates are isolated from caries sites)	– (No evidence)	± (Keratolytic organisms have not been isolated from initial enamel caries but found in deeper cavities)
ii. Presence of necessary bacteria in oral cavity	+ (Bacterial flora capable of producing decalcifying acids always present in the oral cavity and on the tooth surface)	+ (Proteolytic types, such as *Actinomyces bovis*, *Streptococcus mitis* are generally present in the oral cavity)	± (No evidence)
iii. Bacterial antagonists reduce caries	+ (Certain bacterial antagonists, such as iodoacetic acid, penicillin, etc. reduce caries)	– (No evidence)	– (No evidence)

(+) Positive evidence, (±) Doubtful evidence, (–) No/Negative evidence

chemical conditions at enamel-plaque interface may favour outward movement of ions and at other times, the situation may be reversed. This delicate balance of ions is dependent on many factors.

The three important factors responsible for maintaining balance are:

i. pH of plaque.
ii. Calcium and phosphate ion concentration at the interface.
iii. Fluoride ion concentration

If the pH falls below 5 (the critical pH), may be during carbohydrate intake, mineral ions are liberated from the hydroxyapatite crystals of the enamel surface and diffuse into plaque. Within 20 minutes, the salivary buffers neutralize the acid. At this stage, the plaque is supersaturated with ions. Some of the ions are lost and others are deposited onto the enamel. With such repeated episodes, overall demineralization occurs which may lead to caries. For this phenomenon to occur, actual mineral ion

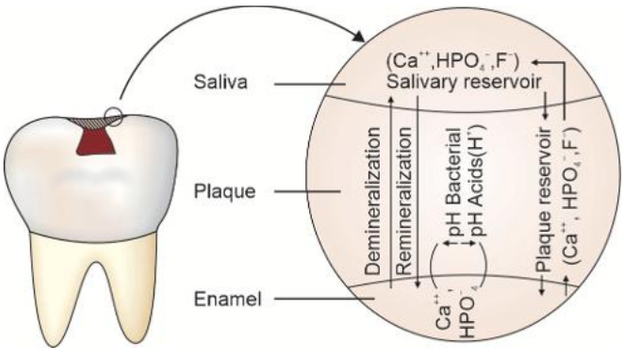


Fig. 3.14: Levine's ionic see-saw mechanism

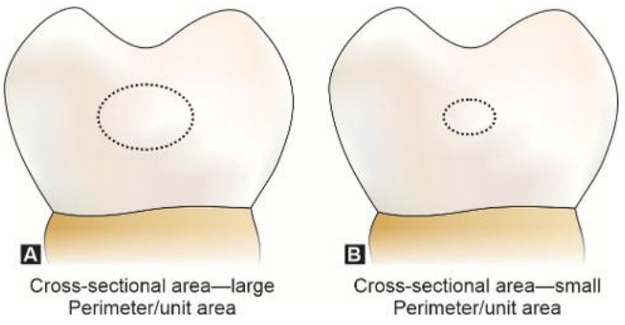


Fig. 3.15: Perimeter/unit area (cross-section): Progression of caries is faster in tooth **(B)** as compared to tooth **(A)**

concentration of saliva is important. If free calcium and phosphate ions are higher in the saliva because of the dietary and other sources, there would be a greater tendency of ions to move from plaque to enamel. Reverse would be true, if ion concentration in saliva is low. Another factor, which plays an important role in 'see-saw' mechanism is fluoride. Fluoride favours movement of ions from plaque to enamel. The initial deposit appears to be in the form of calcium fluoride. Fluoride concentration, as low as 5.0 ppm, can tilt the 'see-saw' in enamel's favour.

j. Bandlish's Theory

Dr. Bandlish (1981) emphasized the role of attrition and plaque in caries etiology. He hypothesized that oral fluids protect the enamel by providing a protective covering. Attrition makes the fissures wider and removes the superficial layer of enamel along with the initial carious lesion, if present. The new layer of enamel becomes protective again with the help of oral fluids. In areas, where the oral fluids cannot reach (e.g. contact areas), enamel cannot be made protective against the carious attack.

Organic acids produced by the bacterial fermentation of the carbohydrate cause surface demineralization of the enamel. At this stage, the bacteria do not penetrate the intact enamel surface. The demineralized products are redeposited into the enamel during the process of remineralization. Caries occurs, only if there is more demineralization and less remineralization.

Caries starts at the contact area, where the protective action of oral fluids is lacking, i.e. the perimeter of the contact area. Smaller the contact area, more is the perimeter/unit area; as the area becomes large, the perimeter/unit area falls. As the contact area becomes more (may be physiologic or with attrition), the perimeter/unit area falls. The effect of acid attack depends upon the perimeter/unit area (Fig. 3.15A, B).

Greater the perimeter, stronger is the attack. As the length of the perimeter/unit area falls with increase in size of the contact area, the carious lesion progresses faster in a smaller contact area, if other conditions are kept constant (Fig. 3.16A to D).

The convexity of the adjoining surfaces is one of the factors, which affects the incidence or intensity of caries. The more the convex surface, the smaller is the contact area—the less the convex surface, the larger is the contact area. When there is contact between two surfaces of varying convexities, the contact area on the more convex surface is smaller. Accordingly, the caries incidence is higher in more convex surfaces. The above mentioned concept of convexities when applied to the caries incidence of different adjacent surfaces seems to be correct except in the case of the contact between the first permanent molar and second deciduous molar. Mesial surface of the first permanent molar is less convex, still this surface is more prone to caries. The reason could be that some caries might have been initiated during the period the first molar remained in contact with the second deciduous molar.

Dr. Bandlish was of the view that meticulous brushing and cleaning reduces caries not by removing plaque but by removing some part of enamel.

It is suggested that gradual recession of active enamel lesion is a result of surface wear. Plaque acts as a reservoir of minerals and with its buffering capabilities helps in maturation of enamel surface, thereby reducing caries.

The approximal surfaces gain the beneficial effect of oral fluids during functional movement of teeth. During the buccolingual movement of teeth, there is more movement at the occlusal border of the contact area and less at the cervical border (less movement nearer the fulcrum). Where there is more movement, there is less incidence of caries; where there is less movement, there is more incidence of caries. Accordingly, the cervical border of the contact area is more prone to caries (Fig. 3.17A).

In case of occlusal caries, caries starts at places where there is least attrition and least plaque, i.e.

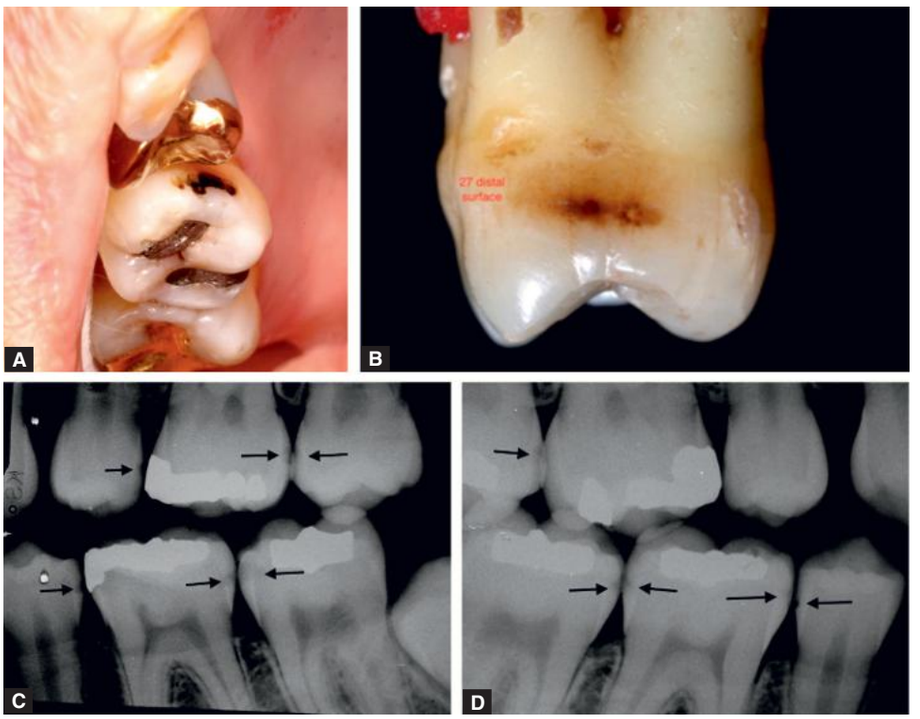


Fig. 3.16: Caries at the contact area

where contact of two or more enamel surfaces occurs like in fissures (Fig. 3.17B) and at cusp tips. He suggested that pits and fissures are not usually full of plaque as the general belief is, but there is only little plaque in the functioning pits and fissures. It is not the depth of a fissure, which is important but the narrowness of the fissure, which makes it more caries prone. Maximum enamel caries occurs in a fissure, where the two walls of a fissure either meet or come close to each other. The plunging action of masticatory forces helps to pump the oral secretions into the occlusal fissure of the teeth. The masticatory forces also help to pump the oral secretions into the region of contact area. Food consumption habits are changing. Since less fibrous and softer food is consumed more, the masticatory forces are reduced. The reduced masticatory forces do not force oral secretions into the unprotected areas thereby making these areas more prone to caries.

According to this concept, attrition and plaque are regarded as defense mechanism against caries.

k. Systemic Theory

The systemic theory highlights the significance of disturbance of oxidant–antioxidant balance in the internal environment of cells that lead to initiation and progression of caries. It states that the acidic environment primarily erodes the enamel and

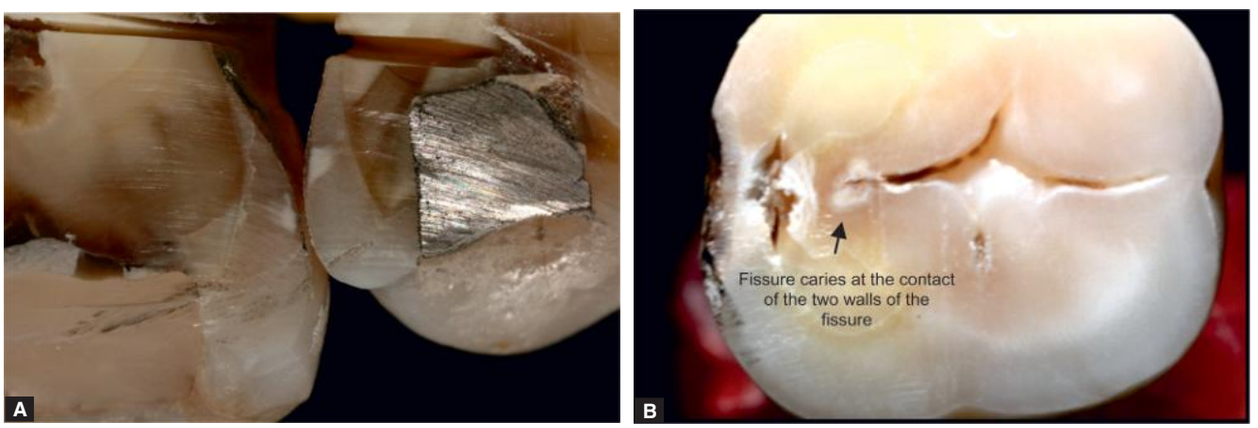


Fig. 3.17: Initial caries at the base of contact area (ground section)

secondarily triggers an inflammatory response in the dentin layer of the tooth. Similar to the process of periodontal disease, where bacterial toxins initiate the body's own matrix metalloproteinases (MMPs) to breakdown the periodontium, the collagen matrix of the dentin is lysed by endogenous collagenase. Essentially, both dental caries stimulated by acid and periodontal disease stimulated by bacterial toxins share a common pathway of oral bacteria initiating a host inflammatory response. This is a significant departure from the concept of the oral bacteria's MMPs creating the dentinal breakdown as recognized in Miller's acidogenic theory.

The systemic theory conceptualizes that all body parts constantly move between state of health and disease; based on neutralization by antioxidants and subsequently the corresponding inflammatory response. A similar reaction takes place in the periodontal tissues and teeth. Nourishment, by either blood supply or fluid flow, replenishes nutrients and antioxidant reserves in cells. The body has mechanisms to regulate hormones and fluid flows. Dentinal fluid flow is regulated by the endocrine portion of the parotid gland, which receives signals from the hypothalamus. Free radicals, specifically reactive oxygen species (ROS), produced in the mitochondria of the hypothalamus increase with elevated blood glucose. Minimizing the effect of free radicals on the hypothalamus with antioxidants can avoid downregulation of the parotid hormone, maintaining centrifugal dentinal fluid flow. Excessive MMPs release causing carious process is prevented, and dentinogenesis is enhanced.

A high-sucrose diet affects the tooth from outside by enabling oral bacteria to produce acid and from inside by reducing the dentinal fluid flow. Antioxidants can shield the dentition by decreasing the effects of acid erosion, by minimizing the effect of ROS in the hypothalamus, and by replenishing tissue inhibitors of metalloproteinases in the dentin.

l. Caries Balance Concept

Featherstone (1999) proposed 'caries balance concept' and hypothesized that the caries is the outcome of disturbance in balance between pathological and protective factors. The caries progress, if the pathological factors overweigh the protective factors.

The pathological factors are:

- Cariogenic/acidogenic micro-organisms
- Quality of fermentable carbohydrates
- Frequency of intake of carbohydrates and their clearance
- Inadequate mineral content and flow of saliva

The protective factors are:

- Adequate mineral contents and flow of saliva
- Supply of fluorides from extrinsic sources (fluoride along with salivary calcium and phosphates provides ingredients for remineralization)
- Effective use of antiplaque

This balance between pathologic factors on one side and protective factors on the other side dynamically changes throughout the day. According to the caries balance concept, caries does not result from a single factor; rather, it is the outcome of the complex interaction of pathologic and protective factors. The risk factors, which may vary with time, must be assessed regularly. The basic concept can be used to treat or arrest/reverse caries in individuals and also at community levels. By checking the pathological factors and increasing the protective factors, one can dream of caries-free dentition.

m. Genetic Theory

Caries has been established as a multifactorial disease, affected by the action and interaction of genetic, environmental, and behavioral factors. Horowitz, et al (1958) investigating 'caries experience rate' observed definite correlation of hereditary to caries susceptibility. The genetic aspect of the host influences factors, such as, host immune response, effect of salivary components and dietary habits that affect the nature of the available substrate upon which the cariogenic organisms grow and multiply. Various authors have confirmed the role of genetics in caries susceptibility; however, a few authors observed no role of environmental factors in caries susceptibility, whereas, genetics play a vital role in individuals' resistance against caries. Many studies provided evidence of genetic contributions, salivary gland functions, structural modification of enamel and varied immune response to cariogenic micro-organisms. Only specific genes have been associated with caries risk, such as genes involved in enamel formation (amelogenin, ameloblastin, tuftelin). The role of proline-rich proteins (PRP-proteins coded by chromosomes) in saliva and other forms of glycoproteins have been evaluated.

The genetic/hereditary involvement of preferences of carbohydrate intake vis-à-vis taste of individuals have also been studied. Studies examining genetically determined taste sensitivity to 6-n-propylthiouracil showed that individuals with low taste sensitivity experience a low caries risk than those with high taste sensitivity. Evaluating genes TAS2R38, TAS1R2 and GNAT3, it was established that taste plays a vital role in caries initiation.

Heritability is the ratio of the genetic component of variance to the total variance of the trait, expressed

as a proportion or percentage. Heritability may vary among populations with different patterns of environmental factors, and may even change with age within a given population. The genetic factors, such as DNA methylation, which is modified by environmental factors and nutrition also to play vital role in the genetic determinant of caries.

n. Autoimmune Theory

Burnet (1959) hypothesized 'forbidden clone' concept, describing caries as the 'autoimmune' disease. He proposed that the normal state of immunological self tolerance might be induced by gene mutations in mesenchymal stem cells. A mutant cell propagates formation of 'forbidden clone'.

These mutant cells synthesize cellular or humoral autoantibodies, which circulate through the body fluid and attack target cells (odontoblasts cells in caries). The theory enables us to understand, how somatic mutations affect large number of target cells at various parts of the body.

Burch and Jackson (1966) later analyzed caries epidemiologic data and suggested that partially mutational genes determine whether a site on a tooth is at risk or not. Since most of the data was based on epidemiologic studies, its authenticity could not be established.

CONTRIBUTORY FACTORS IN DENTAL CARIES

Dental caries is a multifactorial disease. It is reasonable to assume that tremendous variations in caries incidence exist because of number of factors affecting caries. Caries develop because of interaction of four principle factors; the host, the microflora, the diet (substrate) and the time. All the four factors should be favourable, viz. a susceptible host, cariogenic oral flora and a favourable diet for a sufficient length of time for the initiation of caries. Diagrammatically, these factors can be portrayed as four circles overlapping each other. Caries develop (center) when all the factors are favourably acting (Fig. 3.18).

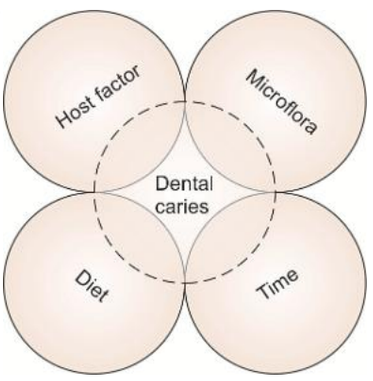


Fig. 3.18: Contributing factors in dental caries

The four factors contributing to caries process are as follows:

1. Host factor
   A. Tooth
      a. Morphology and position in the arch
      b. Chemical nature
   B. Saliva
      a. Composition
      b. pH
      c. Antibacterial properties
      d. Quantity and viscosity
2. Microflora
3. Diet (substrate)
   a. Physical nature
   b. Chemical nature
4. Time

1. Host Factor

A. Tooth

a. ***Morphology and position in the arch:*** Tooth morphology and the position of tooth in the arch are important factors for initiation of caries. Deep pits and fissures make the tooth susceptible to caries because of food impaction and bacterial stagnation. That is why the occlusal surfaces are more prone to caries. The most susceptible teeth are the mandibular first molars amongst the permanent teeth, closely followed by maxillary first molars, then mandibular and maxillary second molars and so on.

Irregularities in the arch form, crowding and overlapping of the teeth also favour the development of caries. Partially impacted third molars are more prone to caries and so are the buccally/lingually placed teeth.

b. ***Chemical nature:*** The inorganic constituents, such as dicalcium phosphate dihydrate, and fluoroapatite, etc. make the enamel resistant to caries attack. It has been established that surface enamel is more caries resistant than the subsurface enamel. The surface enamel has more minerals and organic matter and relatively less water. In addition, certain elements such as fluoride, chloride, zinc, lead, etc. accumulate more on the surface enamel than the subsurface enamel.

With the passage of time, teeth become more resistant to caries because of decrease in permeability and increase in nitrogen and fluoride content. The increase in concentration of fluoride at the subsurface is because of ingestion of fluoride with age. It is hypothesized that under practical limits, higher the fluoride concentration of water, the lower the prevalence of caries.

B. *Saliva*

a. *Composition:* The flow of saliva affects the oral environment vis-à-vis, the dental caries.

The composition of saliva varies considerably. The concentrations of inorganic calcium and phosphorus show considerable variations within resting and the stimulated saliva. Caries prone individuals have low calcium and phosphorus levels. The organic component of saliva has been studied in relation to dental caries. The caries immune persons exhibit greater ammonia content in saliva. The higher ammonia content in saliva retards the plaque formation and neutralizes acid formation. Urea and amino acids have no effect on dental caries. A variety of enzymes, such as amylase and ptyalin, do help in degradation of starches, but their role in caries inhibition is questionable.

b. *pH:* The pH at which the saliva ceases to be saturated with calcium and phosphorus is referred to as the 'critical pH'. Under normal conditions, the critical pH is 5.5. Below this value, the inorganic material of the tooth may dissolve. The normal pH of resting saliva is 6–7. A 'buffer' is a component that tends to maintain a constant pH. A fall in buffer capacity of saliva leads to increase in caries incidence.

c. *Antibacterial properties:* The antibacterial properties of saliva have a definite relation with caries incidence.

Lysozyme activity in saliva has been found to be greater in caries-free children than the caries-susceptible ones. Salivary peroxide is also considered antibacterial; however, no significant correlation has been found between caries resistant and caries susceptible subjects. Antibodies like secretory IgA and IgG against specific bacteria have been reported in human saliva. Patients with immune dysfunction tend to have more caries than normal.

d. *Quantity and viscosity:* The quantity and viscosity of saliva has definite influence on caries incidence. Human beings suffering from decreased flow of saliva or lack of salivary secretions (xerostomia) usually experience increased rate of dental caries (e.g. Sjögren's syndrome, sarcoidosis, diabetes, etc.). However, cariogenic substrate is required for initiation of caries, even if salivary secretions are lacking or minimum.

The caries susceptibility increases in patients following radiation therapy of oral cavity. The decreased salivary flow and subsequently decreased buffering capacity favours the growth of more aciduric yeasts.

Certain drugs influence salivary flow, and in turn result in rampant caries. Drugs, which may lead to xerostomia, include antidepressants and anti-histamines, etc.

Physiological xerostomia occurs in all human beings during sleep. The most important time for plaque removal is before sleeping to avoid caries.

2. Microflora

It is established that micro-organisms are mandatory for caries. The following factors justify the role of bacteria in caries process:

i. Caries do not occur in the absence of micro-organisms.
ii. Caries can occur in animals, even if kept on single type of bacteria.
iii. All oral organisms are not cariogenic; however, majority can be isolated from carious enamel and dentin.

It has been established that caries is a bacterial infection; however, the cariogenicity of micro-organisms is variable. It is also recognised that all acid producing organisms are not cariogenic.

The role of micro-organisms varies at different sites as follows:

a. *Pit and Fissure Caries*

The micro-organisms prevalent in pit and fissure caries are:

- *Streptococcus mutans*
- *Streptococcus sanguis*
- Lactobacilli
- *Actinomyces*
  - Pit and fissure caries have the highest prevalence of micro-organisms.
  - Pits and fissures provide excellent mechanical shelters for micro-organisms.
  - The relative proportion of *Streptococcus mutans* determines the cariogenic potential.
  - Complex communities dominated by filamentous bacteria, such as those in gingival crevice, fail to develop in pit and fissure habitat.
  - The appearance of *Streptococcus mutans* in pit and fissure is usually followed by caries.
  - Sealing the pits and fissures just after tooth eruption may provide resistance to caries.

b. *Smooth Surface Caries*

The micro-organisms prevalent in smooth surface caries are:

- *Streptococcus mutans*
- *Streptococcus salivarius*
  - The proximal enamel surfaces especially immediately gingival to the contact area are the

most susceptible (relatively free from the effects of mastication, tongue movement and salivary flow, etc.) to plaque accumulation.
- Plaque accumulation on these surfaces depends upon oral hygiene, configuration of interdental col, functional/masticatory movements of teeth, etc.
- Rough surfaces (poor quality restoration or a structural defect) facilitate plaque accumulation.
- Proximal surfaces of young children are less favourable habitats for *Streptococcus mutans*.
- Often the gingival aspect of the facial and lingual smooth enamel surface that is supragingival but gingival to the occlusogingival height of contour is neither rubbed by the bolus of food nor cleaned by the tooth brush. Presence of caries in these surfaces is usually indicative of a caries active individual.

c. Root Caries

The micro-organisms prevalent in root caries are:
- *Actinomyces* spp. [*A. viscosus*, *A. odontolyticus*, *A. naeslundi*, *A. israeli*, *A. bovis*, *A. gerensceriae* (*Actinomyces* are the early colonized micro-organisms of root caries)]
- *Streptococci* spp. (*S. mutans*, *S. sanguis*, *S. salivarius*)
- Other filamentous rods.

Caries originating on the root is alarming since:
- Comparatively rapid progression
- Often asymptomatic
- Rapid progress, close to pulp
- Difficult to restore, especially in hidden areas
  - The proximal root surfaces are usually concave especially near the cervical line and remain unaffected by the conventional oral hygiene procedures.
  - The cervical areas, especially when exposed to the oral environment (as a result of gingival recession), favour initiation of caries.
  - The facial and lingual root surfaces (particularly near the cervical line) remain unaffected by oral hygiene measures.

A variety of bacteria contribute to root surface caries. The stage of lesion is important, as it would correspond to the presence of different bacteria. A unique bacterial flora exists on the roots as compared to the crown. Several studies have indicated that the oral microflora gets modified with advancing age, introducing non-oral bacterial species, such as staphylococci and enterobacteria. This might be due to impairment of the immune system. It is established that the root surface is more vulnerable to the demineralization process than enamel because cementum begins to demineralize at pH 6.7, which is higher than the enamel's critical pH of 5.5. Various studies have confirmed the diversity of root caries microbiota. Among gram-positive isolates, *Actinomyces* spp are present in abundance, especially *A. naeslundii*. Other micro-organisms, include, *Streptococcus mutans*, *lactobacilli*, *Bifidobacterium*, *Veillonella* spp and also enterococci. A few authors have also reported staphylococci (mainly *S. haemolytism* and *S. hominis*) from active as well as arrested carious lesions; whereas, *S. epidermidis* reported from active lesions only. *Streptococcus mutans* and *Streptococcus sobrinus* were not associated with active carious lesions; whereas, *Streptococcus sanguis* has been associated with arrested and active lesions both. Recent studies have demonstrated the presence of *Bifidobacterium*, *Scardovia* and *Parascardovia* in initiating root caries along with mutans streptococci, lactobacilli and yeasts. Earlier scientists could identify 20 different strains of gram-positive filamentous organisms (most common—*Rothia dentocariosa*) including *Actinomyces* strains. Other strains resembling *Actinomyces* were also detected. They opined that *Actinomyces* strains could be associated with root surface lesions.

d. Subgingival Area

The subgingival areas harbour the following micro-organisms:
- *Lactobacillus* spp.
- *Streptococcus mutans*
- *Actinomyces naeslundii*
- *Actinomyces viscosus*
- Other filamentous rods
  - Metabolites released from plaque penetrate the epithelial lining of the sulcus.
  - The capillaries dilate and become permeable resulting in the leakage of blood plasma into the tissue.
  - Some metabolites have chemotactic properties that induce infiltration of WBCs into the region.
  - The gingival inflammatory reaction results in release of plasma-like fluid containing immunoglobulin, PMNs, albumins, etc. (These immunologic materials may change characteristics of the adjacent plaque by removing the susceptible organisms.)
  - New niches then become available because of loss of some species and availability of new ones.

Different organisms display some selectivity as regard their preference of tooth surface. *Streptococcus mutans* is considered to be the significant micro-organism playing a vital role in initiation of caries. The main etiological micro-organism in pit and fissure

caries is the *Streptococcus mutans* because of the following properties:

- It ferments mannitol and sorbitol (synthesized insoluble polysaccharide from sucrose)
- Mutans streptococci are lactic acid formers, which easily colonize on tooth surface.
- They are more aciduric than other streptococci.

A few of these properties have also been shown by non-cariogenic strains, such as enterococci, *Streptococcus faecalis*, etc. Two properties, which separate them from other streptococci are (i) acid accumulation by *Streptococcus mutans* is substantially greater than other oral streptococci; (ii) *Streptococcus mutans* contains lysogenic bacteriophage, which has not been isolated from non-cariogenic strains.

e. Deep Dentinal Caries

The flora of deep dentinal caries is different. The predominantly present micro-organisms are lactobacilli, which account for one-third of the oral flora. Certain gram-positive anaerobes and filaments are also present, such as *Eubacterium*, *Actinomyces*, *Bacillus*, etc.

3. Diet (Substrate)

'Diet' refers to the customary food and 'nutrition' means the assimilated portion of diet, which affects the metabolic process of body. Diet has shown to influence caries.

The dietary features responsible for dental caries are:

i. Roughness
ii. Palatability
iii. Eating and drinking pattern
iv. Retention and clearance of diet
v. Age

The following dietary factors play role in caries production.

A. Physical Nature (Raw Food)

It has been established that the physical nature of diet indirectly affects caries. The diet of primitive man consisted of raw food, which led to attrition and also help in cleansing the debris; thereby, less incidence of caries. Modern diet includes refined foods and soft drinks, which helps collection of debris predisposing to more caries. Further, it is observed that the mastication of food reduces the number of micro-organisms. Mechanical rubbing and cleaning definitely has role in caries reduction.

B. Chemical Nature (Nutrients)

The nutrients present in food have definite role in caries process.

a. *Carbohydrates:* Carbohydrates are important component of food. They are principally a source of energy; help in synthesis of fatty acids and amino acids. They are also a part of structure of biologically important materials, such as glycolipids, glycoproteins, nucleic acids and heparin.

The nutritionally important carbohydrates are:

- Glucose (occurs mostly in combined form and is found in nearly all foods)
- Sucrose (most familiar and prominent sugar in domestic use)
- Lactose (unique to mammals and found in milk)
- Starch (energy stored in most plants and seeds)
- Glycogen (animal equivalent of starch)

Various authors have documented the role of sugar and sugar-containing food items in the development of caries. The individuals having habit of consuming excess sugar in routine food, experience more caries. Many authors in their respective studies have confirmed the association of sugar and caries.

The famous Vipeholm Study was conducted on 436 adult inmates in a mental institution. The relationship between carbohydrate intake and caries was evaluated, commonly referred to as '*Carbohydrate Study I*'.

The inference of study was:

- The consumption of sugar can increase caries activity.
- The risk of caries is greater, if the sugar is consumed in a form that will be retained on the surfaces of the teeth.
- The risk of sugar increasing caries activity is greatest, if sugar is consumed between meals.
- Upon withdrawal of sugar-rich foods, the increased caries activity rapidly disappears.
- Caries lesion may continue to appear despite the avoidance of refined sugar and maximum restrictions of natural sugars and dietary carbohydrates.
- The increase in caries activity varies widely from one person to another.
- The increase in clearance time of the sugar increases the caries activity.

The study showed that the physical form of carbohydrates was much more important in cariogenicity than was the total amount of sugar ingested.

The decalcification potential of a food is equivalent to retention of food product coupled with their potential of acid production.

The factors affecting cariogenicity of carbohydrates are divided into two groups:

i. *Factors related to food products*
   - Types of carbohydrates

- Quantity/concentration of carbohydrates
- Stickiness
- Chewing/swallowing

ii. *Factors related to acid production*

- Frequency of intake
- Oral clearance
- Timings of intake
- Behavioural characteristics

i. ***Factors related to food products***

- *Types of carbohydrates:* The major dietary source of carbohydrate is starch, which are slowly fermented in oral cavity by salivary amylase. Starch products when combined with sucrose for sweetness are more cariogenic than sucrose alone. Most of these products, such as cookies, breads, etc. are consumed as between meal snacks, increasing their cariogenic contribution. The soft drinks contain starch hydrolysate (glucose group), which may have even higher caries potential than processed and heated starch.
- *Quantity/concentration of carbohydrates:* The quantity and concentration of carbohydrates are important for development of caries; however, no linear relationship has been observed.
  Various authors have opined that plaque pH was lower after rinsing with 10–50% concentration of sucrose.
- *Stickiness:* The stickiness of food items enhances the retention time of sugars, resulting in a prolonged fall in pH, subsequently increased incidence of dental caries. High retention rates have been observed for products, such as sweet biscuits, potato chips, etc.
- *Chewing/swallowing:* Chewing results in increased contact of food items with the tooth surfaces, subsequently increased incidence of dental caries as compared to swallowing. However, acid in fruits juices and soft drinks may decrease the oral pH. The fall in pH may enhance fermentation of carbohydrates, leading to demineralization and caries.

ii. ***Factors related to acid production***

- *Frequency of intake:* The frequency of intake of sugar may have a significant effect as regard to cariogenicity of the diet.
  The frequency of intake gains importance as caries is regarded as the outcome of balance of demineralization and remineralization. The duration of the decrease in pH after intake of a cariogenic food is important.
- *Oral clearance:* Oral clearance implies clearance of carbohydrates from the oral cavity. A few authors have observed that frequency of intake might not be related to the development of caries, but the time the sugars remain available to micro-organisms in the oral cavity is significant. Oral clearance of carbohydrates depends on metabolism by micro-organisms, degradation of carbohydrates by plaque and salivary enzymes, and the flow of saliva. A short clearance time reduces the length of time the sugar remain available for acid production by the bacteria. The clearance rate is of great importance for elderly population, where medically induced low salivary secretion may lead to root caries.
- *Timing of intake:* The timings of sugar intake, especially before sleep time is important feature for caries development. The low salivary flow during sleep decreases oral clearance of the sugars and increases the contact time, subsequently increasing the cariogenic potential of sugar and other food items.
- *Behavioural characteristics:* The food habits vary from individual-to-individual. A variety of factors, such as availability, cost, traditions, taste, emotions and even advertisements, etc. govern our choice of food.

b. *Fats:* It has been established that dietary fats have limiting effect on dental caries.

Medium chain fatty acids and their salts have antibacterial property at low pH; however, their mechanism is not well defined.

Certain fatty acids have exhibited antimicrobial effects and have been shown to inhibit carbohydrate metabolism in plaque.

The anticariogenic effects of dietary fats are attributed to:

- May alter the effect of diet by replacing carbohydrate in snacks.
- May form a physical barrier on tooth surface and prevent demineralization.
- Enhance clearance of carbohydrates from the oral cavity.

c. *Proteins:* It has been established that protein deficiency during dental development leads to smaller teeth, delayed eruption and a greater susceptibility to caries. Arginine rich peptides and vitamin B_6 (pyridoxine) may have protective effect against caries. Metabolism of these amino acids by plaque bacteria may inhibit/offset the production of demineralizing acids, resulting in increase in pH level within the plaque.

The mechanism by which protein deprivation induces caries susceptibility is not established; however, following factors may contribute:

- Reduction in salivary flow; subsequently, the buffering capacity.
- Reduction in remineralization and antibacterial capacity.
- Tooth morphology may get modified/altered.
- Decrease in immune response (salivary IgA level falls).

d. *Vitamins*

i. *Vitamin A:* Vitamin A is concerned primarily with the process of differentiation of epithelial cells. Odontogenic epithelium fails to undergo normal histodifferentiation and morphodifferentiation in vitamin A deficiency and results in an increased rate of cell proliferation. The ameloblasts also fail to differentiate properly.

The association of dental caries with vitamin A deficiency has not been documented exhaustively; only a few studies have reported higher caries in post-eruptive vitamin A deficiency. This may be due to changes in salivary gland functions rather than changes in dental tissues. A few authors have observed that cod liver oil (rich source of vitamin A) supplements were able to decrease the dental caries susceptibility.

ii. *Vitamin B complex:* Vitamin B complex is essential as growth factor for oral bacteria. These vitamins serve as components in coenzyme involved in glycolysis process of bacterial metabolism. In animal studies, vitamin B complex deficiencies have lead to disturbances in maintenance of tooth supporting tissues; however, their role in caries initiation is not documented.

Niacin appears to promote dental caries via stimulation of cariogenic oral flora. Vitamin B_6 (pyridoxine) has shown some anticariogenic action as it promotes growth of non-cariogenic microflora in oral cavity thereby hampering growth of cariogenic micro-organisms. Vitamin B_7 (biotin) is essential nutrient for all acid producing micro-organisms and may play a role in microbiology of caries.

iii. *Vitamin C:* It is established that vitamin C is more involved in maintenance of gingival health than caries.

iv. *Vitamin D:* Vitamin D enhances absorption of calcium and phosphate from the food. The inorganic component increases the strength of teeth and their ability to fight demineralization. Deficiency of vitamin D, especially during developmental stage, results in delayed eruption of teeth.

The beneficial effects of vitamin D are:

- Affects calcium metabolism.
- Induces cathelicidin, an antimicrobial peptide, which has the potential to attack oral bacteria causing dental caries.

From the inference of various studies, it is concluded as follows:

- Taking vitamin D supplements results in substantial reduction of dental caries in children.
- Taking vitamin D supplements had no effect on caries, especially in older age.
- It has also been established that geographic location and sun exposure are related to dental caries. People living closer to the equator with greater amounts of sun exposure are less likely to develop dental caries. Use of vitamin D appears to be a better option for reducing dental caries than fluoridation of community water supplies, as there are many additional health benefits of vitamin D and a number of adverse effects of water fluoridation, such as fluorosis (mottling) of teeth and bones.

v. *Vitamin K:* Vitamin K may be effective in reducing caries, since it is known to reduce enzymatic activity in carbohydrate degradation cycle; however, a few authors have observed that deficiency of vitamin K did not promote dental caries incidence.

e. *Minerals*

i. *Calcium and phosphorus:* Calcium (Ca) and phosphorus (P) are essential components of saliva and are required for remineralizing the demineralized tooth structure. The exact mechanism of action of phosphate in caries reduction is unclear; however, evidence suggests local effect in oral cavity rather than the systemic. Calcium and phosphate influence as:

- Ability of phosphate to reduce the rate of dissolution of hydroxyapatite of enamel.
- Supersaturated solutions of phosphates can redeposit calcium phosphate, particularly in areas of enamel that have been particularly demineralized.
- Phosphates act to buffer organic acids formed by fermentation of plaque micro-organisms.
- Phosphate act to desorb proteins from the enamel surfaces thereby modifying acquired pellicle.

Phosphates have shown significant cariostatic activity, largely due to local action on tooth. No correlation between cariostatic activity of phosphates and its water solubility/buffering capacity/chelating ability have been observed.

Adhesiveness or retentiveness of phosphate is important. The compounds, which are retentive on surfaces and permit slow release of phosphate appear to be most effective. Phosphate combined with fluoride have an additive effect on control of dental caries; however, mechanism of action of both is

different. Phosphates because of their detergent properties can interfere with adherence of plaque bacteria to enamel surface and can also reduce bacterial growth.

Lynch (2004) studied the anticaries effects of calcium glycerophosphate, which may act as follows:

- *Effects on enamel:* Monofluorophosphate (MFP) and calcium glycerophosphate (CaGP) in a definite ratio may affect the acid solubility of hydroxyapatite. Various studies have suggested that calcium glycol phosphate interacted directly with outer layer of hydroxyapatite.
- *Buffering of plaque pH:* It is established that 0.25–1.0% calcium glycerophosphate can effectively reduce the subsequent drop in plaque pH.
- *Modification in plaque metabolism:* A few studies observed reduction in mean plaque scores after rinsing with glycerophosphate solutions.

ii. *Sodium and potassium:* Salts of sodium (Na) and potassium (K) are extensively studied as regard their effects on caries. Their individual effect on caries is not established; however, potassium intake showed statistically significant correlation between caries prone individuals and caries.

iii. *Trace elements:* Trace elements are a part of food as good as any other nutrient. Various components of trace elements are:

- *Fluoride:* It has been hypothesized that pre-eruptive exposure to fluoride inhibits caries (fluoride may get incorporated into the developing enamel hydroxyapatite crystal and thus reducing enamel solubility).

  Fluorides act by reducing the enamel solubility by pre-eruptive incorporation of fluoride into the hydroxyapatite crystals. Flourides promote remineralization; inhibit demineralization of early caries and inhibit glycolysis (the process by which cariogenic bacteria metabolize fermentable carbohydrate).
- *Zinc:* Role of zinc in dental caries could not be authenticated. Various studies have observed zinc as cariogenic and also as cariostatic.

  Different concentrations of zinc control growth of *Streptococcus mutans*, initial plaque formation and inhibition of acid production.
- *Copper:* It has been established that copper reduces the acidogenicity of the dental plaque. Copper inhibits various metabolic enzymes through oxidation and forms the insoluble copper-potassium salts on tooth surface, thereby increasing its acid resistance. Its topical application to the teeth inhibit acid production by plaque and produce lower *Streptococcus mutans* count.
- *Molybdenum:* A few authors have observed a decrease in caries prevalence with high intake of molybdenum. Most of the animal studies have established caries reduction with molybdenum.
- *Selenium:* Most of the animal studies have observed that selenium has damaging effect on teeth as far as caries experience is concerned; however, it could not be established in human studies.
- *Vanadium:* Many studies have reported an inverse relationship between vanadium concentration in water and dental caries.
- *Magnesium:* The animal studies have observed increase of caries by magnesium; while epidemiologic studies suggested the opposite. High magnesium is usually present with high calcium and it is difficult to prove any effect of magnesium on teeth without excluding calcium.
- *Manganese:* It is hypothesized that high levels of manganese in both saliva and plaque cause increase in caries incidence.
- *Sulfur:* The deficiencies of sulfur can have deleterious effect on tooth development or caries, could not be established.
- *Chromium:* The role of chromium in caries inhibition is not clear; however, lower level of chromium in oral environment affects growth and metabolism of bacteria.
- *Cobalt:* The role of cobalt in caries initiation/progression could not be established.
- *Iodine:* The anticaries effect of iodine is documented in animal studies. A few authors have observed that iodine has an additive effect on reducing caries when used with fluoride.
- *Iron:* Iron has an antibacterial effect especially on *Streptococcus mutans* and it also interferes with biofilm formation. Caries inhibiting effect of iron might be due to its ability to inhibit the H-ATPase of *Streptococcus mutans*, affecting its acidogenicity and aciduricity. Iron also interferes with sucrose metabolism, leading to reduction in extracellular polysaccharides production. Its definite role in caries has not been established.
- *Nickel:* Nickel may have minor effect on caries reduction.
- *Silicon:* Topical application of silicon on animals produced no caries inhibiting effect. Silicon might incorporate into apatite crystals of enamel leading to increase in their solubility.
- *Lithium:* The effect of lithium in caries reduction is not established.
- *Titanium:* It has been hypothesized that titanium may enhance the cariostatic effect of fluoride on systemic administration. Titanium is considered an effective cariostatic agent both topically and systemically.

- *Lead:* No correlation was observed with concentration of lead in enamel and caries.
- *Cadmium:* The role of cadmium in caries initiation/progress is not established.
- *Strontium:* Strontium causes mottling of enamel; however, its role in caries initiation/progress could not be established.

4. Time

The frequency to which teeth are exposed to cariogenic environments affects the likelihood of caries development. After intake of meals/snacks containing sugars, the bacteria in the oral cavity metabolize them, resulting in production of acids; the subsequent decrease in pH leads to dissolution of inorganic content of the tooth. It takes around two hours to return the pH to normal and remineralize the tooth surface through the buffering capacity of saliva. Since teeth are vulnerable during these periods of acidic environments, the development of dental caries relies on the frequency of acid production vis-à-vis sugar intake.

HISTOPATHOLOGY OF DENTAL CARIES

Mature human dental enamel has a high degree of mineralization making structural studies somewhat more difficult than other oral tissues.

There are several methods used to study dental caries. (Polarized light is mostly used.) The polarized light is more sensitive for contrasting normal from carious surface than other techniques.

The normal histological aspects of tooth tissues is described in brief so as to understand the pathological deviations.

A. Enamel

Enamel is an extremely hard and highly mineralized tissue covering the anatomical crowns of teeth.

The chemical constituents of enamel are:

i. Inorganic matter (95–97%)

Principal constituents:	
Calcium	34.5–37.1%
Phosphorus	17.1–18%
Carbonate	2–2.8%
Magnesium	0.2–0.6%
Trace elements:	
Sodium	0.2%
Potassium	0.05%
Iron	0.02%
Chlorine	0.30%
Fluoride	0.02%

ii. Organic matter (0.2–0.8%), mainly glycoproteins, carbohydrate matrix, Type I collagen

iii. Water (1.2–4%)

iv. Rest is oxygen.

The structure elements of enamel are:

a. ***Enamel rods:*** Enamel rods are the organizational units of enamel. These are polygonal, long, slender prisms having irregular five or six sides with diameter of 4.0 mm. The enamel pattern is described as cylindrical rods embedded in the interrod enamel. The apatite crystals are arranged approximately parallel to the long axis of the rods; deviate about 65° from this axis as they fan out into the tails of the prisms.

b. ***Prism sheath:*** It is the least calcified portion of the enamel rod, which is richer in organic substance than rest of the rods.

c. ***Interprismatic substance:*** Less than 1.0 μm in width, relatively softer, glossy and homogeneous, interprismatic substance is present between the enamel prisms.

d. ***The striae of Retzius:*** They represent the poorly calcified portion of enamel, extending from the dentinoenamel junction towards the tooth surface. Their meeting point at enamel surface forms grooves encircling the tooth known as Perikymata.

e. ***Enamel lamellae:*** These are thin, leaf-like structures that extend from the enamel surface towards the DEJ (rarely extending into dentin); lamellae consist of mainly organic material with little mineral content.

f. ***Enamel tufts:*** These are the hypocalcified enamel rods and interprismatic substances that originate at DEJ and extends into the enamel for about 1/3rd–1/5th of its total thickness.

g. ***Enamel spindles:*** The spindle-like structures, projecting from DEJ into enamel (approximately 100 μm).

The direction of the odontoblastic processes and spindles in the enamel corresponds to the original direction of the ameloblasts and is at right angle to the surface of dentinoenamel junction.

h. ***Hunter and Schreger bands:*** The Hunter and Schreger bands are the optical phenomena produced by the changes in direction between adjacent groups of rods. These are the dark and light alternating zones that can be reversed by altering the direction of incident illumination.

i. ***Enamel membranes:*** These are of two types:
   - *Developmental:* Primary and secondary enamel cuticles
   - *Acquired:* Pellicle, materia alba and plaque

B. Dentin

Dentin is a calcified connective tissue, penetrated by definitely arranged small canals containing protoplasmic processes.

The chemical constituents of dentin are:

i. Inorganic matter (61.0–73.0%)

Principal constituents:

Calcium	33.0–35.4%
Phosphorus	16.0–18.0%
Carbonate	3.8–4.6%
Magnesium	0.9–1.1%

Trace elements:

Sodium	0.20%
Potassium	0.07%
Iron	0.02%
Fluorine	0.02%

ii. Organic matter (20.3–22.4%)

iii. Water 10.8–15.7%

iv. Oxygen (remaining part).

The structure elements of dentin are:

a. ***Dentinal tubules:*** These are minute S-shaped tubules radiating perpendicularly from pulp chamber and extending to the outer surface of dentin. These tubules end perpendicular to the dentinoenamel junction or dentinocemental junction.

b. ***Dentin matrix:*** The dentin matrix is homogeneous and an elastic substance, containing dentinal tubules. The matrix consists mainly of collagenous fibrils, embedded in an organic cementing medium.

c. ***Sheaths of Neumann:*** A deeply stained area around the dentinal tubules, named as sheath of Neumann, is the portion of dentinal matrix forming the immediate walls of dentinal tubule.

d. ***Interglobular dentin:*** These are areas of imperfectly calcified dentinal matrix formed due to incomplete fusion of calcospherites during dentinal development.

e. ***Tomes' granular layer:*** A granular area seen just below the surface of root dentin in ground section under transmitted light.

f. ***Incremental lines of von Ebner:*** The lines run at right angles to the dentinal tubules and mark the normal rhythmic, linear pattern of dentin deposition in an inward and rootward direction.

g. ***Contour lines of Owen:*** These are incremental lines of dentin produced by the disturbance in calcification of that layer at the time of its development.

h. ***Mantle dentin:*** It is the first formed dentin (approximately 15 μm wide) in the crown close to the dentinoenamel junction; bound by a zone of interglobular dentin.

i. ***Predentin:*** It is the first deposited layer of unmineralized matrix (thickness varies 10–50 μ) and lines the innermost (pulpal) portion.

j. ***Dead tracts:*** Dentinal areas characterized by degenerated odontoblastic processes; may result from injury caused by caries, attrition, erosion or cavity preparation.
They appear black in transmitted light and white in reflected light.

k. ***Secondary dentin:*** It is the dentin formed after the completion of root formation and represents the slow and continuous deposition of dentin.

l. ***Transparent dentin/sclerotic dentin:*** It is formed due to vital reaction of dentin to irritation, e.g. caries; the odontoblastic processes undergo fatty degeneration and become a barrier for progression of caries.

m. ***Tomes' fibers:*** These are the protoplasmic projections of the odontoblasts found in the dentinal tubules.

C. Dentinoenamel Junction

Dentinoenamel junction (DEJ) has a three-dimensional scalloped appearance along the dentin–enamel interface.

Under scanning electron microscope (SEM), the junction represents a series of ridges, which increases adherence between dentin and enamel. This ridging is more developed in coronal dentin where occlusal stresses are greatest. The junction between dentin and enamel is often scalloped with convexities directed towards dentin and concavities towards enamel. This region has high proportion of interprismatic substance, considerable crossing and branching of dentinal tubules, enamel spindles and persistence of some of the organic substance (dentinoenamel cuticle).

The width of the DEJ is dependent on the intratooth location; 12 μm width at the occlusal side and 6.2 μm at cervical side, which might be related to its function.

D. Cementum

There are two types of cementum according to origin and structure:

i. *Acellular cementum:* The cementum present on the coronal and middle part of root, which is without any cellular content.

ii. *Cellular cementum:* The first formed cementum present in the apical third of the root.

The structure elements of cementum are:

a. ***The matrix:*** It consists of fine fibers, connected to each other by a calcified connecting substance.

b. ***Incremental layers of cementum (lines of Salter):*** The periods of rest following the activity period result in an incremental appearance of cementum in sections. The width between these layers is

responsible for the thickness of cementum on a particular root surface.

c. ***Cementocytes and lacunae:*** The cells of cementum (called cementocytes), present in the irregular spaces known as lacunae (not present in acellular cementum).

d. ***Sharpey's fibers:*** These are the fibers of periodontal ligament, embedded in cementum and serve to attach the tooth to the surrounding bone.

E. Cementoenamel Junction

The interrelationship of cementoenamel junction (CEJ) has been classified into four different categories: Enamel overlapped by cementum (60–65%), edge-to-edge contact of cementum and enamel (30%), and gap between cementum and enamel (5–10%) that exposes the dentin. The fourth category (rare), wherein enamel overlapping cementum is difficult to explain from an embryological standpoint, because cementum formation begins after enamel formation is completed.

The enamel prisms, near the CEJ, are directed approximately 90° to the long axis of the tooth. In longitudinal sections, the prisms assume parallel wave-like courses in a plane parallel to the cut surface. Occasionally, the enamel rods near the CEJ may lie in an arc, the peripheral end of which is directed towards the root apex.

Prisms directed apically and occlusoangularly may occur at various places along the CEJ of the same tooth.

The CEJ is an important site where gingival fibers attach to a healthy tooth. In young adults, the CEJ of permanent teeth is protected by the gingival tissues. However, with increasing age and continuous passive eruption, environment make it vulnerable to pathological changes, such as root caries, abrasion/erosion, etc.

F. Cementodentinal Junction

The cementodentinal junction (CDJ) forms a biological and structural link between cementum and dentin. It is 2.0–3.0 μm wide, full of proteoglycans (mucopolysaccharides and core proteins). The fibrils intermingling between cementum and dentin occur only in a few places.

The mineral contents of this zone are less as compared to either cementum or dentin; therefore, the mechanical properties are inferior. The inferior mechanical properties facilitate reducing the stress concentrations between the two mechanically dissimilar tissues.

This region of altered composition facilitates biological and biomechanical functions in addition to providing the structural integrity.

The proteoglycans mainly contribute to its structural integrity under wet conditions.

The distribution of individual cementum types relate to the past functional demands. In initial acellular cementogenesis, the first collagen of root dentin is deposited about 1.0 mm from the basal lamina of the epithelial sheath. The intervening gap contains dentinal ground substance and only a few sparsely distributed collagen fibrils. At the initial dentin mineralization, the mineralization does not involve this surface layer, which mineralizes later to form the CDJ. With a breakdown of the sheath, proteoglycans on which primitive principal fibers are attached, appear in the unmineralized superficial dentin layer.

In the cellular cementum, only a small number of intrinsic or extrinsic fibers penetrate the CDJ to reach the dentin surface. In addition, the CDJ is more irregular in cellular cementum than acellular cementum. The overall function of CDJ could aid absorption and distribution of occlusal loads between cementum and dentin before release into alveolar bone.

The effects of damage on cemental surface may be seen as areas of hypo- or hypermineralization and areas of resorption due to alterations in organic and inorganic components. The bacterial toxin traverses cementum, reaches CDJ and causes destruction of the unmineralized collagen, which is present in abundance at CDJ.

The area of destruction where collagen is lost appears as vacuoles under the microscope and is known as 'pathologic granules'. The destruction of fibres at CDJ could cause weakening of the CDJ in periodontitis affected teeth. More so, the tissues of a weakened junction may not withstand mechanical periodontal procedures resulting in complete removal of cementum, thus exposing dentin.

I. CARIES OF ENAMEL

The stages involved in the progress of enamel caries are: (i) The organic framework is thickened; (ii) the process invading the prisms making them more acid resistant while the prism sheaths turn black; (iii) the prism sheaths disappear while the isolated prisms remain; and (iv) a necrotic mass replacing the enamel structure.

The carious process may vary slightly, depending upon the occurrence of the lesion on smooth surfaces or in pits or fissures. Accordingly, caries in enamel is discussed under two headings:

a. Smooth surface caries

b. Pit and fissure caries

a. Smooth Surface Caries

On smooth surface of tooth, the smooth chalky white area of decalcification manifests as the initial macroscopic evidence of incipient caries. The white spot enamel surface is hard and shiny and cannot be distinguished from the surface of adjacent sound enamel using the sharp explorer point. These surface lesions may appear brownish and are described as brown spots. This largely depends on the degree of exogenous material absorbed by the porous region and amount of water in the structure. The demineralization occurs due to increase in porosity of enamel and is related to relative refractive index (RI) of enamel (1.62), air (1.0) and water (1.33). In subsurface lesion, the pores are filled with a watery medium. As the pores are air dried, air replaces the water. As the differences of refractive indices are increased, the lesion looks more opaque and hence becomes more evident.

Enamel rods run parallel or even converge slightly from the entrance of cavity to dentinoenamel junction. Thus, caries penetrate along the axis of enamel rods. In longitudinal ground sections, defect appears as cone shaped with the apex of the triangle directed towards the dentin and base towards the surface of the tooth (Fig. 3.19A, B)

b. Pit and Fissure Caries

The occlusal fissures are deep invaginations of enamel surface. The carious lesion usually initiates at both sides of the fissure wall rather than at the base which can be visualized as chalkiness, yellow, brown or black discoloration. Further, early dentin involvement frequently occurs as the enamel in the bottom of the pit or fissure may be very thin; however, some pits and fissures are shallow and have relatively thick layer of enamel covering their base. In both types, the enamel rods flare laterally in the bottom of the pits and fissures. The carious lesion characteristically forms a triangular or cone-shaped defect with its apex at the outer surface and its base toward the dentino-enamel junction as it follows the direction of the enamel rods. The general shape of the lesion is just the opposite of that occurring on smooth surfaces as the enamel rods diverge from the surface to the dentinoenamel junction (Fig. 3.19A,B). The early stages of caries is shown in Fig. 3.20.

Zones of Enamel Caries

The following four zones of enamel caries have been recognized, beginning from the enamel surface towards the dentin (Fig. 3.21).

Zone 1: The Surface Layer

The surface layer ranges from 20 to 100 µm. It is thinner in active lesions and thicker in inactive ones. Quantitative studies of the surface layer indicate that partial demineralization equivalent to about 1.0–10% loss of mineral salts has taken place and the pore volume of the surface zone is less than 5.0%.

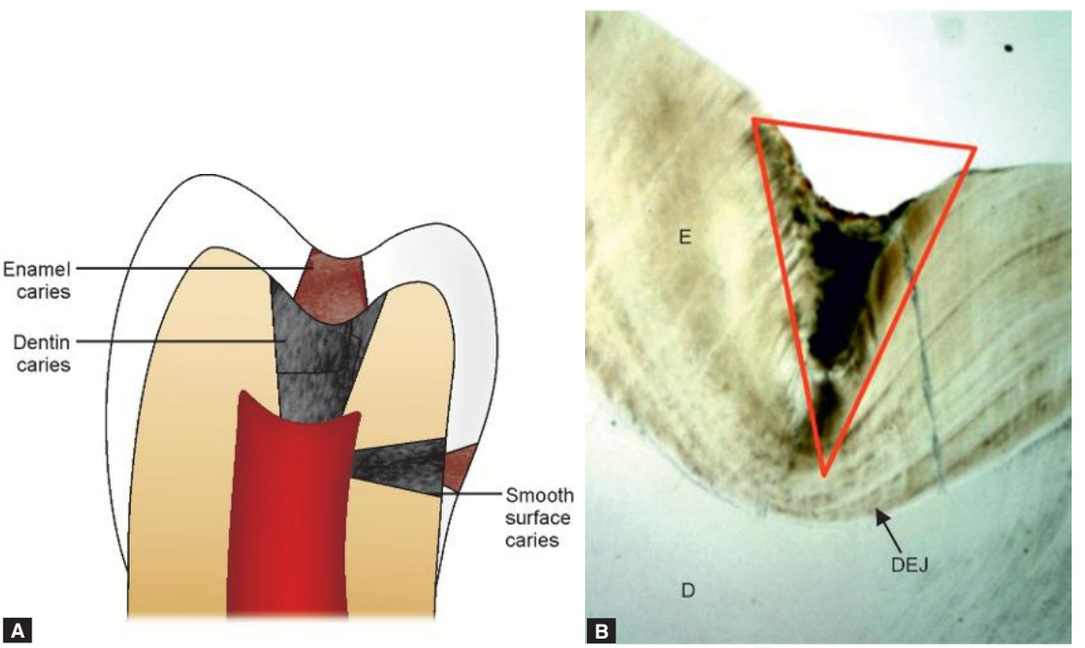


Fig. 3.19: (A) Progress of caries in the pit and fissure and on smooth surface (diagrammatic); **(B)** Longitudinal section of tooth showing enamel caries; triangular in shape with apex towards dentinoenamel junction (DEJ) and base towards enamel (E: Enamel; D:Dentin; DEJ: Dentinoenamel junction)

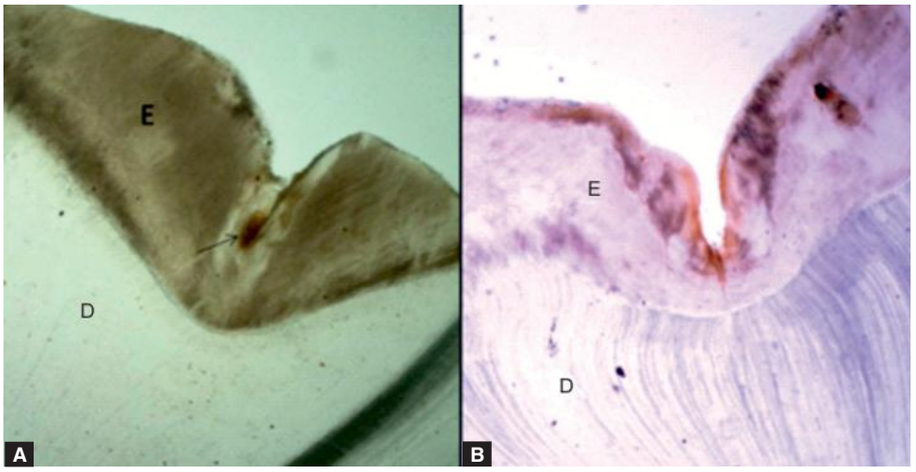


Fig. 3.20: Longitudinal ground section showing stages of caries progression: **(A)** Occlusal pit caries; **(B)** Early enamel caries along the fissure (E: Enamel; D: Dentin)

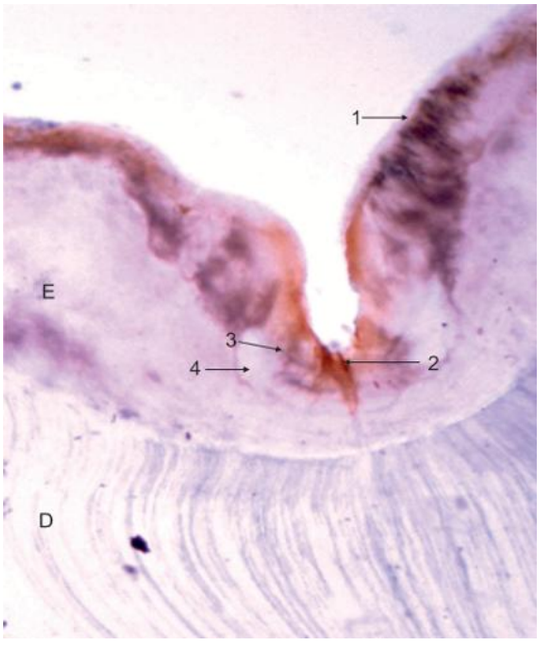


Fig. 3.21: Longitudinal ground section of tooth depicting zones of enamel caries. 1. Surface zone; 2. Body of lesion; 3.Dark zone; 4.Translucent zone. (E: Enamel; D: Dentin)

After imbibing with a medium like water whose refractive index is less than that of the enamel, although the porous subsurface zone is seen to be positively birefringent, the surface zone retains a negative birefringence. This relatively unaffected surface zone is also identifiable on microradiographs as sharply demarcated from the underlying radiolucent regions of the lesion. Thus the surface zone, when examined by polarizing microscope, has been defined as the zone of negative birefringence, superficial to the positively birefringent body of lesion. It is the most unaffected zone due to the great resistance of the surface zone to decalcification because of the greater degree of mineralization of the surface (hypermineralized surface) and increased concentration of fluoride in the surface enamel where the pore volume is less than 5.0%.

- Surface layer relatively immune to caries due to hypermineralization (higher fluoride contact and saliva contact)
- Pore volume less than body of the lesion
- Partial demineralization

Zone 2: The Body of Lesion

This zone lies below the relatively unaffected surface layer. It is the largest portion of the incipient lesion in a demineralizing phase. In polarized light, the zone shows pore volume of 5.0% in spaces near the periphery, to 25% in the center of the intact lesion.

When a longitudinal ground section is examined in quinoline with transmitted light, the body of lesion appears relatively translucent compared with sound enamel. This zone, unlike normal enamel, is positively birefringent when examined in water, denoting significant degree of mineral loss. However, the striae of Retzius within this region are well marked and, therefore, enhanced in contrast to sound enamel. The interprismatic areas provide access to the rod (prism) cores, which are then preferentially attacked. Bacteria may be present in this zone, if the pore size is large enough to permit their entry.

- In demineralization phase, it is the longest zone. [Dark zone (zone 3) is largest in remineralization phase.]
- Large pores (pore volume 5.0–25%).
- Usually bacteria enter pores.

- Striae of Retzius well marked (Striae of Retzius is the initial point of entry of caries into rod/prism core of enamel.)

Zone 3: The Dark Zone

The dark zone lies beneath the body of the lesion and adjacent/superficial to the translucent zone. This zone appears dark brown in ground sections examined by transmitted light after imbibition with quinoline. Polarized light studies have shown that the dark zone has pore volume of 2.0–4.0%.

Dark zone shows positive birefringence in contrast to the negative birefringence of sound enamel. Hence it is often referred to as the positive zone. These effects have shown to be due to the presence of small pores in the zone besides the relative large pores that are present in the translucent zone. Therefore, when a ground section is examined in a mounting medium (quinoline), the relatively larger molecules of medium are unable to penetrate the micropore system of the dark zone. Since the micropores remain filled with air or vapor, light is scattered on passing through the zone, causing brown discoloration of the dark zone. In a similar manner, the presence of a medium or low refractive index within micropore system is responsible for the reversal of birefringence when examined in polarized light. There is some speculation that the dark zone is not really a stage in the sequence of the breakdown of enamel; rather, the dark zone may be formed by deposition of ions into an area previously only containing large pores. It must be remembered that caries is an episodic disease with alternating phases of demineralization and remineralization. Experimental remineralization has demonstrated increases in the size of the dark zone at the expense of the body of the lesion. There is also a loss of crystalline structure in the dark zone, suggestive of the process of demineralization and remineralization. The size of the dark zone is probably an indication of the amount of remineralization that has recently occurred.

- Many tiny pores block light transmission (smaller air/vapor-filled pores make the region opaque).
- Loss of crystalline structures suggests process of demineralization and the remineralization occurring in this zone.

Zone 4: The Translucent Zone

The translucent zone lies at the base of enamel lesion and is recognized as the zone of alteration from normal enamel. Only half of lesions demonstrate this zone at their advancing front, which is seen only when longitudinal ground section is examined in clearing agent having refractive index (RI) identical to that of enamel, e.g. quinoline (RI = 1.62). The name refers to its structureless appearance and examined with polarized light. The spaces or pores created in the tissue in this stage of enamel caries are located at prism boundaries and other junctional sites. In this zone, the pores or voids form along the enamel prism (rod) boundaries, presumably because of the ease of hydrogen ion penetration during the carious process. When these boundary area voids are filled with quinoline solution, which has the same RI as enamel, the features of the area disappear. By means of polarized light, it has been shown that this zone is slightly more porous than sound enamel, having pore volume of 1.0% compared with 0.1% in sound enamel. There is no evidence of protein loss in the translucent zone.

- Deepest zone, represent advancing front of enamel caries.
- Pores/voids form along the enamel prism (rod) boundaries (due to easy hydrogen penetration).
- Appears structureless when seen with polarized light (hence translucent).

II. CARIES OF DENTIN

Caries in dentin begins along the dentinoenamel junction with the rapid involvement of greater number of dentinal tubules. The dentinal tubules act as tract leading to the dental pulp. The micro-organisms may also travel along these tubules. According to Arnold, et al., the demineralization of dentin causes a patho-morphological reaction of the dentin–pulp complex in response to the carious attack, resulting in peritubular, intratubular and reactive dentin deposition within the dentinal tubules and pulp chamber. The caries progressing into dentin have been shown differently by earlier authors.

Arnold et al. classified carious dentin into following six zones:

a. Softened dentin
b. Demineralized dentin
c. Bacterial invasion into the tubules
d. Translucency zone of dead tracts
e. Hypermineralized translucent zone
f. Secondary dentin

Marshall, et al classified carious dentin into two different layers—caries-infected dentin (CID) and caries-affected dentin (CAD).

Caries-infected dentin is a superficial layer, full of pathogenic bacteria and bacterial byproducts. It is characterized by destruction of mineral matrix or crystal. This stage could not be remineralized due to high levels of decalcification or demineralization. On histological evaluation of this layer, degraded

collagen fibers with disruption of cross-links were observed. The discontinuation of cross-links indicated an irreversible stage of collagen, or denatured collagen, and thus demonstrating the unremineralizable nature of this layer.

Caries-affected dentin constituting the inner zone is partly but highly demineralized. This zone is free of bacteria and is remineralizable due to the presence of sound collagen fibers and presence of hydroxyapatite crystals. This zone should ideally be preserved during tooth preparation.

Active Caries of Dentin

The carious process after invading enamel, reaches dentinoenamel junction and spread laterally. The spread into dentin is orthrograde due to its histology and mineralization. The progress and speed of caries in dentin is characterized by following five zones starting from pulp to dentinoenamel junction (Fig. 3.22).

Zone 1: Zone of Fatty Degeneration of Tomes' Fibers

This zone is present at the most pulpal end of dentinal caries. Due to the irritation from bacterial products, the fats get deposited in the Tomes' fibers. When stained with Sudan red; two types of fat lipids are evident in this zone—one from the bacterial products and other derived from the dissolution of the intratubular dentin. The peritubular and intertubular dentin remain intact. The term 'fatty degeneration' is misnomer, as there is no fatty degeneration of Tomes' fibers; however, there is deposition of fat in Tomes' fibers.

- Characterized by presence of a layer of fat globules (stains red with Sudan red stain)
- Fat layer leads to impermeability of the dentinal tubules—trying to prevent further invasion of carious lesion.
- Also favours sclerosis of dentin in zone 2.

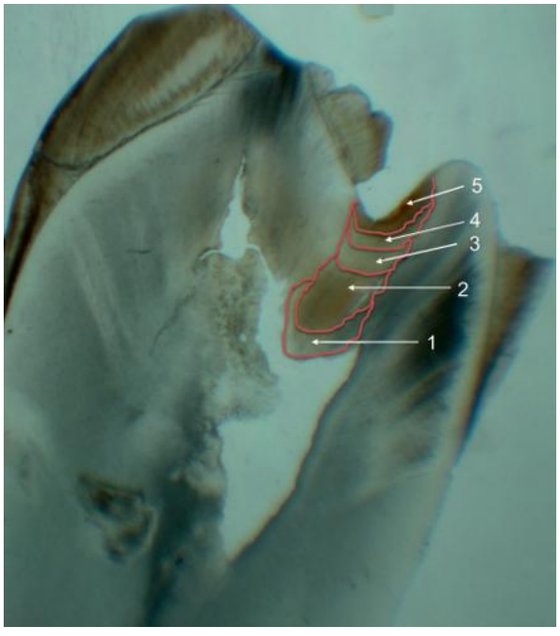


Fig. 3.22: Longitudinal ground section of tooth showing zones of dentin caries. 1. Zone of fatty degeneration of Tomes' fibers, 2. Zone of dentinal sclerosis, 3. Zone of decalcification, 4. Zone of bacterial invasion, 5. Zone of decomposed dentin

Zone 2: Zone of Dentinal Sclerosis

This zone is characterized by deposition of mineral salts in the dentinal tubules leading to the calcification of intratubular dentin. It is a reaction of vital dentinal tubules and vital pulp so as to seal off the healthy tooth structure from the diseased one and also prevent further penetration of the micro-organisms into healthy tooth. The formation of sclerotic dentin is minimal in rapidly advancing caries and is more evident in chronic caries. When viewed under transmitted light, it is transparent; whereas, in reflected light, it is dark. The peritubular and intertubular dentin are evident in this layer. The translucent dentinal tubules will be localized precisely beneath the demineralized enamel surface and will appear even before the enamel demineralization has reached the dentinoenamel junction.

- Calcification of dentinal tubules as a reaction of vital pulp to carious invasion so as to prevent invasion of micro-organisms.
- Sclerotic dentin appears white in transmitted light.

Zone 3: Zone of Decalcification of Dentin

This is a narrow zone preceding the bacterial invasion. The peritubular dentin is absent in some areas of the inner part of this zone; whereas, it is intact in the outer part. In intertubular dentin, the collagen is normally present and the inorganic contents are mostly dissolved. The decalcification commences from the lateral walls of the tubules leading to their distension as they get filled with microbes in next zone. In early stage of caries, when only a few of the tubules are involved, micro-organisms may be found penetrating these tubules before there is any clinical evidence of carious process—'the pioneer bacteria'. Examination of individual tubules has shown almost pure form of bacteria, i.e. one tubule may be filled with cocci, while the adjacent tubules may contain only bacilli or spirochetes.

- Initially, only walls of dentinal tubules get decalcified.
- Bacteria present in individual dentinal tubules are in pure form (i.e. either completely cocci or completely bacilli and not in mixed variety).
- Presence of 'pioneer bacteria' before there is any clinical evidence of caries.

Zone 4: Zone of Bacterial Invasion

In this zone, bacterial invasion of the decalcified, intact dentin occurs. Abundance of bacteria is present in the enlarged lumens of dentinal tubules or intratubular dentin. In the early stages, acidogenic bacteria are in abundance and in deeper layer, proteolytic bacteria replace the acidogenic bacteria. This zone supports the hypothesis that initiation is by acidogenic bacteria and the progress is by proteolytic ones. Acidogenic bacteria utilize carbohydrates while proteolytic bacteria utilize dentinal protein for their metabolism. Even the intertubular dentin is invaded by bacteria in advanced carious lesions. Inner peritubular dentin is completely absent and the outer peritubular dentin is present only in parts. Inorganic content of peritubular dentin is absent but not entirely; whereas, collagen fibers are present.

- Presence of micro-organisms; in early stages only acidogenic bacteria present and in progressive stage, proteolytic bacteria predominate.

Zone 5: Zone of Decomposed or Necrotic Dentin

Due to the bacterial invasion of dentinal tubules in the zone 4, there is an increase in diameter of these tubules; which are further broken down and coalesced so as to form the ovoid areas of destruction—known as liquefaction foci of Miller. Also thickening and swelling of sheath of Neumann, may sometimes be noted at irregular intervals in this zone. The liquefaction foci are present parallel to the course of dentinal tubules. It is filled with necrotic debris that tends to expand with time resulting in compression and distortion of adjacent dentinal tubules thus further increasing the carious lesions. Inner peritubular dentin is absent completely and the outer peritubular dentin is faintly present. Inorganic content of intertubular dentin is almost entirely absent; whereas, collagen is present in early stages. This zone is most distinct in advanced dentinal caries.

- No recognizable structure seen.
- Collagen and minerals may be absent.
- Correct number of bacteria dispersed in decomposed granular matter.
- Most superficial zones of dentin caries.

III. ROOT CARIES

The study of histopathology of root caries is divided into caries of cementum and caries of root dentin.

a. Cemental Caries

The cemental caries is different from the enamel and dentin caries as the cementum is composed of more organic and less mineralized components as compared to enamel and dentin.

i. *Active Caries*

The active cemental caries has three patterns:

- *Pattern I:* This pattern is found most predominantly and is due to the high intensity acid attack in the oral cavity after exposure to sugars.
  It is characterized by the uniform demineralization of both cementum and its underlying dentin. The incremental lines of cementum are accentuated in these lesions. Cementum has higher degree of residual mineral components than the affected dentin.
- *Pattern II:* This pattern is generally present in the abraded, eroded or denuded surfaces either clinically or microscopically.
  This pattern is evident in roots exposed to mild acid attack. Initially, small clefts transversing the cementum and extending into dentin are present. These first get enlarged and then filled with microbes subsequently. These enlarged clefts with microbes form the microcavities exposing the peripheral dentin. Eventually, the exposed peripheral dentin gets covered by microbes and cementum borders get undermined by the micro-organisms. The destruction of cementum occurred stepwise along the fractures following incremental lines of cementum. In this pattern, no immediate bacterial invasion of peripheral dentin occurs.
- *Pattern III:* This pattern is least distinct and is found in the areas where demineralization and remineralization processes are going simultaneously, e.g. in the persons having high exposure to fluorides. In this pattern, a surface remineralized layer of 10–30 µm thickness is present. It is interrupted over well-defined areas of advanced demineralization. Uniform demineralization of cementum and peripheral dentin is the characteristic feature of these lesions.

b. Root Dentin Caries

i. *Initial Lesions*

The multiple clefts present in the cementum enter the dentin along with their microorganisms. These clefts spread laterally towards the pulp, fuse with each other to form microcavities filled with microbes (Fig. 3.23). As they enlarge further, the bacteria moved deeper towards the open dentinal tubules. A distinct clearly demarcated layer of complete demineralization can be distinguished from underlying intact dentin.The spread of caries occurs along the dentinal tubules through intertubular dentin (Fig. 3.24).

ii. *Advanced Lesions*

The advanced lesions of root dentin have shown various areas of radiolucency and radiopacity with two predominating patterns:

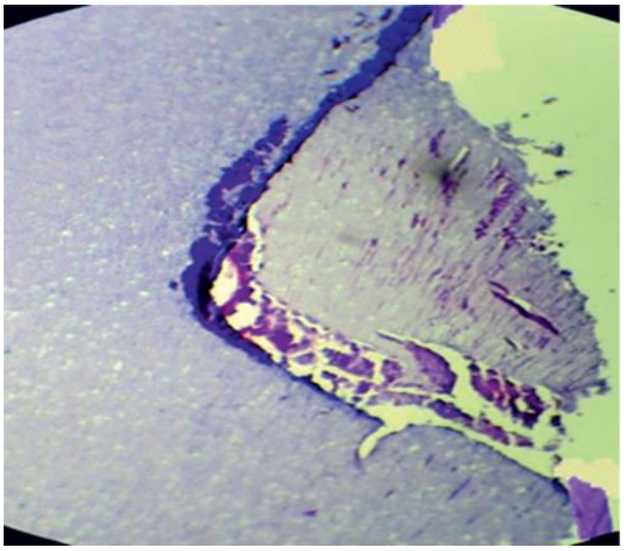


Fig. 3.23: Microphotograph of carious cementum: Small clefts traversing the cementum and extending into dentin

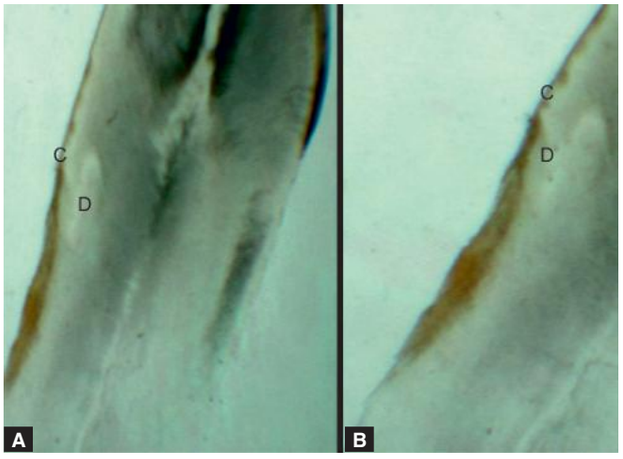


Fig. 3.24: Longitudinal ground section of tooth showing root caries (caries of cementum). **(A)** Initial lesion, **(B)** Considerable dentin involvement (C: Cementum; D: Dentin)

- *First pattern:* In this pattern, the radiolucent areas appeared as enlarged lesions with variable extent and depth along the exposed root surface. Some of the radiolucent areas are covered by the radiodense layer while others are not. In addition, radiodense layers were observed bordering the radiolucent areas over the inner dentin.
- *Second pattern:* In this pattern, only a single radiolucent saucer-shaped area extending into dentin is present. Occasionally, this area is covered by the radiodense layer in parts. Rarely, a band-like radiopaque layer is present around the radiolucent lesion.

Histopathologically, the radiolucent and radiodense areas show following features:

Radiolucent area:

- Intertubular dentin is almost completely demineralized.
- Dentinal tubules and their lateral branches are occupied by micro-organisms.
- Peritubular dentin of those tubules is absent.

Radiodense area:

- Accumulations of small needle-like crystals throughout them.
- Massive spread of microbes is present.
- Microbes penetrating the dentinal tubules spread laterally, along the (parallel to) collagen fibers and reach adjacent tubules.
- Subsequently, the micro-organisms totally engulf the dentinal tubules.
- They, then form a radiodense band bordering the demineralized areas present towards inner dentin.

iii. *Arrested Lesions*

There are two types of arrested lesions in root dentin:

- *Truly/definitely arrested lesions:* These lesions are completely mineralized, with a distinct zone of tubular sclerosis. Dentinal tubules appear as solid rods embedded in partially demineralized dentin. The occlusion of dentinal tubules occurs with fine granular crystals.
- *Intermediate/mixed lesions:* These lesions show a mineralized surface layer and underneath this layer, a mineralization front advancing into demineralization zone is present.

Both lesions have following common features:

- Intertubular dentin is completely mineralized till surface.
- Their surface is covered with microbes.
- Dentinal tubules near the surface are packed with ghosts of micro-organisms.

ARRESTED CARIES

Arrested caries implies absolute slow progress or no progress of active carious lesions. The conditions how the arrest of an active lesion occurs are not very clear. An arrested dentinal lesion differs from an active lesion by its darker pigmentation, absence of visible bacteria within the tubule and impermeability to dyes and isotopes. Three layers are identified in an arrested lesion:

i. A narrow surface layer, brown in color and leathery in consistency.
ii. The widest layer in the lesion, dark brown in color and hard in consistency.
iii. Sclerotic layer is very hard, white and often harder than normal dentin.

In the most superficial layer, occasional bacteria are present. The peritubular zone is partially absent. In contrast to the surface layer of active lesions, this area is more dense and seems to contain more mineral. The dense condensation at the surface forms a narrow, homogenous calcified border approximately 2.0–5.0 μm in width. The main bulk of arrested carious lesion consists of deeply pigmented and hard dentin. Almost all the tubules in the area contain bacteria. Under this layer, there occurs a bacteria-free zone of almost normal appearance and corresponds to deep calcified layer.

The main features of the deep sclerotic zone are the highly calcified contents of the tubules. No bacteria are observed in either the sclerotic zone or the normal dentin below.

One of the major characteristics of an arrested lesion is its higher degree of mineralization. The possible ways by which the minerals get accumulated are:

- Remineralization of the surface layer (calcium and phosphate are absorbed from saliva).
- Reprecipitation of dissolved apatite (the minerals dissolved by bacterial acids in the upper demineralized layer reprecipitate to form large apatite crystals).
- Sclerosis by intratubular calcification and obliteration of tubules of deeper layers. (The minerals are mediated by odontoblastic processes through blood supply.)

The other characteristic is the change in bacterial morphology. In the arrested lesions, the bacterial bodies are coalesced into homogenous masses, which could be the sign of degeneration and disintegration. The bacterial bodies present in the arrested lesion are probably degenerated and non-viable. The third characteristic is the deep pigmentation. The degenerated bacteria or the degradation products of their proteins and nucleic acids are the possible source of pigmentation. Bacteria are very rarely observed in the deeper portions of both the layers of arrested lesion. The underlying affected but bacteria-free layers can remineralize.

Histopathology

The histopathology of arrested caries is described in three zones:

i. The inner barrier (the surface formed as a result of tubular sclerosis in the area between carious and sound dentin).
ii. The outer barrier (formed by a highly mineralized surface).
iii. The mineralized zone (an area extending from the outer barrier towards the root canal in the demineralized dentin).

i. *The inner barrier* is the surface formed due to tubular sclerosis. The degree of tubular sclerosis is directly related to the arrest of progress of caries. The permeability of dentin is reduced, thereby the diffusion of acids and proteolytic enzymes from carious dentin do not reach pulp. The formation of inner barrier usually depends upon the integrity of odontoblast layer in the pulp. However, arrested lesions also occur in the absence of functional pulp.
ii. *The outer barrier* blocks the diffusion of bacterial metabolism products into dentin. Certain crystals, independent from odontoblastic processes are found in dentinal tubules near the surface.
iii. *The mineralized zone* or the remineralization depends upon the degree of bacterial invasion and the integrity of tissue. The zone of bacterial penetration is separated from the zone of advancing demineralization by a mineralized band located below the surface. It is established that the conversion from an active to arrested lesion is associated with the influx of ions thus preventing further demineralization of dentin. However, it is not clear whether the conversion is based only on mineral deposition at surface of lesion or complete remineralization of the body of the lesion.

SECONDARY CARIES

The caries around a restoration is termed as 'secondary caries' and also known as 'recurrent caries'. The caries may be present at surface enamel surrounding the restoration or extend underneath it along the margins. The caries on the opposing tooth surface in contact with the restoration is termed as 'contact caries'. The main etiological factor for secondary caries is marginal leakage around the restorations. It is most unfortunate that all restorations leak, the degree, however, varies. It has been observed that in cases where the width of the marginal defect is less than 50 μm, the risk for secondary caries is low. Other reasons, which may (though less likely) predispose to secondary caries is ditching or fracture of the restoration at the edges. Secondary caries around silver amalgam, cast restoration and ceramic restoration is shown in Fig. 3.25A–C. A rough surface of the filling coupled with poor oral hygiene may retain more plaque, subsequently the acids, which can diffuse into the microspaces adjacent to the cavity walls. Also, an improper cavity preparation where fissures are left untouched renders the tooth susceptible to secondary caries (Fig. 3.26B). The formation of secondary carious lesions may be higher in unpolished than in polished enamel surfaces.

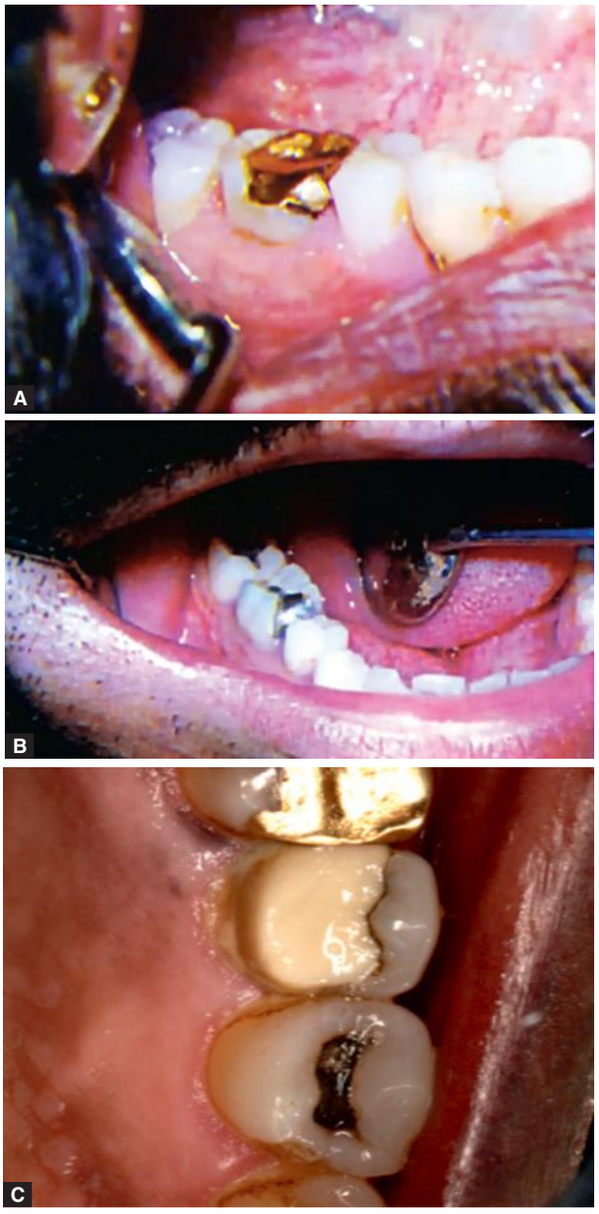


Fig. 3.25: Secondary caries around: **(A)** Cast restoration; **(B)** Amalgam restoration; **(C)** Ceramic restoration

Radiologically, a thin radiolucent line is seen under the restoration (Fig. 3.27A, B).

Epidemiology

A few epidemiological studies evaluation secondary caries are:

- Goldberg, et al. (1981) reported 19.8% occlusal, 9.7% mesial, 14.9% distal, 11.8% lingual and 23.9% buccal secondary caries. In total, 17.7% surfaces showed recurrent caries in 914 restored teeth.
- Eriksen, et al. (1986) in their study of 1694 Class II restorations, found 5.5% to be affected by recurrent caries; 4.9% recurrent caries was observed with amalgam restorations; 4.3% with composite restorations and 1.3% with gold restorations.
- Espelid and Tveit (1991) classified secondary caries as S-1 (initial caries lesion characterized by discoloration only), S-2 (lesions characterized by softness and/or cavitation in enamel), and S-3

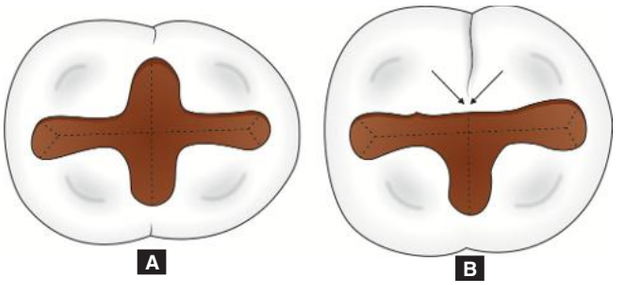


Fig. 3.26: Effect of cavity preparation: **(A)** Proper cavity preparation; **(B)** Improper extension of cavity can predispose to secondary caries (arrows)

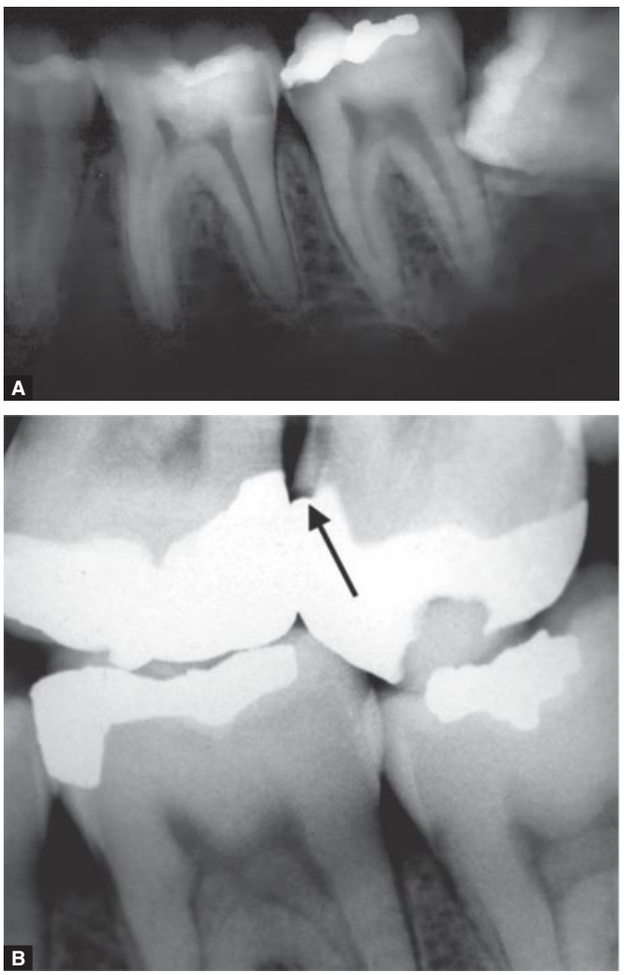


Fig. 3.27: Secondary caries (radiographic): **(A)** A thin radiolucent line seen under the restoration (mandibular molar); **(B)** A thin radiolucent area under the restoration (maxillary molar)

(lesions with cavitation on the root surfaces). Their observations were 89.7% S-3 lesions, 58.5% S-2 lesions and 73.3% S-1 lesions. They, however, did not specify the status of the occlusal caries.

- Pimenta et al. (1995) in their study on amalgam restorations showed that 47.16% surfaces were non-ditched with caries while 54.94% were non-ditched without caries. Further, 58.82% were ditched with caries and 41.18% were ditched without caries.
- Mjor (1998) in his study after examining secondary caries at different locations under different materials observed secondary caries in 3.8% Class I amalgam restorations, 0.4% Class I composites, 4.3% glass-ionomer and 4.3% unspecified. In Class II restorations, amalgam showed 90% gingival caries, 5% occlusal and 10% other sites. Similarly, composite and glass-ionomers showed 75% and 80% gingival, 8% and 10% occlusal and 15% and 10% on other surfaces.

Microbiology

Secondary caries, like primary caries, is believed to be caused by action of acids produced by bacteria. In a way, secondary caries resembles pit and fissure caries. It requires an environment simulating pits and fissures where a storehouse for nutrients is available and micro-organisms can survive and colonize.

The three common organisms seen to be associated with secondary caries are mutans streptococci, Lactobacilli and *Actinomyces viscosus*. However, no single species can be ascertained in its cause and the number and type of organisms vary with different lesions. The presence of significant numbers of these bacteria in dentin of secondary caries indicates their role in the formation and development of secondary carious lesions.

Predominance of streptococci and other types of gram +ve cocci have also been observed in failed restorations. Mutans streptococci and lactobacilli have been found to increase in significant numbers in teeth restored with amalgams with marginal defects wider than 40 μm. The restorative material definitely affects the activities of these micro-organisms. The type of restorative material influences qualitatively and quantitatively the microflora in the lesion; e.g. gamma-2 free amalgams can inhibit the metabolic activity of micro-organisms due to release of copper. Release of fluorides from the restorative materials may also have protective effects.

The micro-organisms found in the secondary dentinal lesions are of same nature as encountered in advanced dentinal lesions or of root dentinal lesions. Dentin is decalcified at pH 5.7 or below and most of the micro-organisms are capable of lowering the pH below this value. Though mutans streptococci and lactobacilli are cariogenic, other less acidogenic bacteria could also be involved in the causation of recurrent dentinal caries.

Histopathology

Secondary caries may reveal an 'outer lesion' or a 'cavity wall lesion' or both. When a tooth is restored, the surrounding tooth structure is considered in two planes—the outer tooth surface and the cavity wall. The 'outer lesion' refers to one which forms on the outer surface of the tooth; whereas, the 'wall lesion' occurs on the walls of the cavity (Fig. 3.28). The outer lesion is considered as a result of primary attack; whereas, wall lesion is because of leakage at the tooth-restoration interface. The penetration of hydrogen ions plays a key role in the process of demineralization. The hydrogen ions may penetrate along the microspaces. (The diffusion, capillary forces and differences in electrical potential between the tooth surface and the filling can lead to penetration of hydrogen ions into the microspaces.)

The progress of outer lesion results in a cavity on the tooth surface adjoining the restoration; whereas, wall lesion results in deeper cavity along the cavity walls. The outer lesion and a greater part of the wall lesion are bordered by a dark zone. In the outer lesions, lesions revealing more than 5.0% spaces are higher than lesions revealing less than 5.0% spaces, (approximate 4.2:1.43). On the other hand, the opposite is true for the cavity wall lesions, i.e. lesions revealing less than 5.0% spaces are higher than lesions revealing less than 5.0% spaces, (approximate ratio 3: 1). Lesions that correspond to less than 5.0% of spaces are visible only by polarized light; whereas, those corresponding to more than 5.0% spaces are observable because of their radiolucent appearance also.

In the dentin, the wall lesion shows a phenomenon indicative of demineralization, as the conventional carious process.

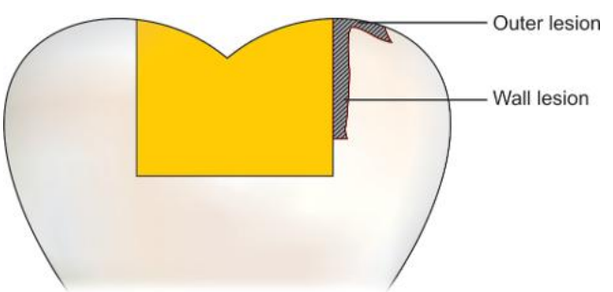


Fig. 3.28: Secondary caries around a restoration: Outer lesion and wall lesion

The types of variations reported in the cavity wall lesions of dentin are:

i. Superficial demineralization of the cavity wall.
ii. Subsurface demineralization without visible changes of the wall surface.
iii. Subsurface demineralization with increased radiopacity of the surface layer (*the radiopacity may be the result of reprecipitation of mineral ions from the underlying demineralized area or because of the deposition of mineral ions from amalgam, especially tin ions, in the surface of the dentin wall*).
iv. Alternating zones of increased radiopacity and radiolucency, as observed towards the cavity:
   - A narrow layer of highly increased radiopacity.
   - A narrow layer of highly increased radiolucency.
   - A narrow layer representing same degree of radiopacity as the intact dentin.
   - A broad radiolucent dentin
   - A broad layer of increased radiopacity.

(Release of ions from a cement base or from amalgam material may have contributed to the radiopacity of layer 1, but the other layers may be a result of the alternations of demineralization and reprecipitation).

Caries in the dentin follows dentinal tubules; while in the cementum, caries proceeds preferentially along the Sharpey's fibers.

Differentiating Secondary Caries with Residual Caries

It is emphasized that residual caries left during cavity preparation must be differentiated from secondary caries. In case of residual caries, the demineralized areas are non-uniformly divided; whereas, in secondary caries, the area displays a fairly constant width in its course along the curvature of the cavity wall. It is very difficult, rather not possible to accept that the well-defined uniform width of demineralization in secondary caries might be due to incorrect excavation of caries or cavity preparation.

The wall lesions of secondary caries and the damage caused to cavity walls during preparation procedures (burns) should also be differential diagnosed. Wall lesions appear translucent while burns appear opaque under ordinary light. Wall lesions show gray fluorescence of low intensity under ultraviolet light while burns show white fluorescence of high intensity. Microradiographically, the wall lesions are radiolucent while burns are radiopaque.

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